DOUBLING IN GENERALIZED POLYGONS: LARGE INCIDENCE STRUCTURES OF DIAMETER 3, 4 AND 6

DRAFT FOR DISCUSSION — AUTHORS TO BE SETTLED

ABSTRACT. Let $n_b(\Delta, D)$ be the largest number of elements (points plus lines) of an incidence structure in which every point lies on at most Δ lines, every line has at most Δ points, and any two elements are at distance at most D in the incidence graph. For $D \in \{3, 4, 6\}$ and $\Delta - 1$ a prime power the classical generalized D-gons of order $(\Delta - 1, \Delta - 1)$ attain the Moore bound. For the remaining degrees we adjoin new points and new lines to a classical generalized polygon. The main tool is Delorme's *doubling* of a set of elements, with levels, which we recast as an operation on incidence structures. We show that in a generalized n-gon, doubling a set of elements keeps the diameter at most n if and only if every connected component of that set is a tree of diameter at most $n - 2$. For $D = 3$ we add four *hats* to any projective plane of order $q \geq 3$ and obtain $2(q^2 + q + 6)$ elements at $\Delta = q + 2$; in $\mathrm{PG}(2, q)$ a correlation completes it to a regular structure. For $D = 4$ we show that two elliptic quadrics through a point x of $\mathrm{W}(q)$ form a weighted 2-ovoid of $\mathrm{AS}(q) = P(\mathrm{W}(q), x)$, q odd, which gives $(q+2)$-regular structures with $2q^2(q+2)$ elements; we also treat copies of lines ("shadows") and give a short proof that two orbits of a Singer subgroup form a 2-ovoid of $\mathrm{W}(q)$. For $D = 6$ we extend Delorme's doubling of the split Cayley hexagon $\mathrm{H}(q)$ along a Hermitian spread, for every odd q, by an explicit set of $q^2(q^2 - 1)$ lines that meets the counting bound, and obtain $2(q+1)(q^4 + q^2 + 1) + 2q^4 + 4q^3 + 4q + 10$ elements at $\Delta = q + 2$, together with regular versions.

1. INTRODUCTION

Let $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ be a finite incidence structure, and let $\Gamma(\mathcal{S})$ be its incidence graph. Every bipartite graph with a chosen bipartition is the incidence graph of an incidence structure. So the *bipartite degree/diameter problem* asks for the largest number of elements of an incidence structure with given maximum degree and given diameter. Write

$$n_b(\Delta, D) = \max\big\{\, |\mathcal{P}| + |\mathcal{L}| \;:\; \Delta(\mathcal{S}) \leq \Delta,\ \operatorname{diam} \mathcal{S} \leq D \,\big\},$$

where $\Delta(\mathcal{S})$ is the largest number of lines on a point or points on a line, and $\operatorname{diam} \mathcal{S}$ is the diameter of $\Gamma(\mathcal{S})$. Counting the elements at distance $0, 1, \ldots, D - 1$ from a flag gives the bipartite Moore bound

$$n_b(\Delta, D) \leq M_b(\Delta, D) = 2\big(1 + (\Delta - 1) + \cdots + (\Delta - 1)^{D-1}\big).$$

Equality holds exactly for the incidence graphs of generalized D-gons of order $(\Delta - 1, \Delta - 1)$. By the theorem of Feit and Higman [11] this forces $D \in \{3, 4, 6\}$ when $\Delta \geq 3$ (an octagon of order (s, s) would need $2s^2$ to be a square). Projective planes, generalized quadrangles and generalized hexagons of order (q, q) are known when q is a prime power, and for no other q. So the open cases with $D \in \{3, 4, 6\}$ are the degrees Δ for which $\Delta - 1$ is not a prime power. In the usual range $\Delta \leq 16$ these are $\Delta \in \{7, 11, 13, 15, 16\}$.

Our constructions all have the same shape. We start from a classical generalized polygon $\mathcal{S}$ of order (q, q) or $(q - 1, q + 1)$, with $q + 1 < \Delta$, and adjoin new points and new lines with prescribed incidences. The incidence graph appears only at the end, as the object whose order and diameter we count.

Results.

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- *Doubling* (Section 3). We describe Delorme's doubling with levels [7, 8] as an operation $\mathcal{S} \mapsto \mathcal{S}_{\mathcal{A}}^{(m)}$ on incidence structures, prove Delorme's diameter theorem and its converse (Theorem 3.2), and show (Lemma 3.3 and Corollary 3.4) that in a generalized n-gon the doubling has diameter at most n if and only if each component of the doubled set is a tree of diameter at most $n-2$. This explains why diameter 3 needs extra "patch" incidences and why diameters 4 and 6 do not.
- *Diameter* 3 (Section 4). A fourth hat in any projective plane of order $q \geq 3$ (Theorem 4.2), completing the construction of Araujo-Pardo and López [2]: $n_b(q+2, 3) \geq 2(q^2+q+6)$, with a regular version in $\mathrm{PG}(2, q)$.
- *Diameter* 4 (Section 5). A weighted 2-ovoid of $\mathrm{AS}(q)$ formed by two elliptic quadrics (Theorem 5.6); clones and shadow lines in any generalized quadrangle (Theorem 5.4); a 2-ovoid of $\mathrm{W}(q)$ from a Singer subgroup (Proposition 5.8); regular structures of every degree $\Delta \geq q+1$ (Corollary 5.10).
- *Diameter* 6 (Section 6). Delorme's doubling of $\mathrm{H}(q)$ along a Hermitian spread, extended by an explicit exact set of $q^2(q^2-1)$ "whisker" lines for every odd q (Theorem 6.5), and regular versions (Theorem 6.12).
- *Orders* (Section 7). Doubling gains at most about $2q^{D-2}$ elements at $\Delta = q+2$; the constructions reach this order of magnitude for $D = 4$ and attain the counting bound for $D = 6$.

2. Notation and preliminaries

2.1. **Incidence structures and distance.** An *incidence structure* $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ consists of disjoint sets $\mathcal{P}$ of *points* and $\mathcal{L}$ of *lines* and a symmetric relation $\mathrm{I} \subseteq (\mathcal{P} \times \mathcal{L}) \cup (\mathcal{L} \times \mathcal{P})$. Points and lines are *elements*. We do not assume that two points lie on at most one line: some of our structures have repeated lines. For a set $\mathcal{A}$ of elements, the *substructure induced on* $\mathcal{A}$ is $(\mathcal{A} \cap \mathcal{P}, \mathcal{A} \cap \mathcal{L}, \mathrm{I})$; its connected components are the components of the subgraph of $\Gamma(\mathcal{S})$ induced on $\mathcal{A}$. We say that $\mathcal{A}$ is a *tree* if this subgraph is a tree.

All distances are taken in $\Gamma(\mathcal{S})$. Two points are at even distance, a point and a line at odd distance. Two distinct points are at distance 2 exactly when they are collinear, and a point p and a line L are at distance 3 exactly when $p \not\mathrel{\mathrm{I}} L$ and some point of L is collinear with p. An element is *isolated* if it is incident with nothing.

Lemma 2.1 (Diameter in geometric terms)**.** *Let $\mathcal{S}$ be an incidence structure without isolated elements and let $D \geq 2$.*

(i) *If D is even, then* $\operatorname{diam}\mathcal{S} \leq D$ *if and only if* $\mathrm{d}(p, L) \leq D-1$ *for every anti-flag (p, L).*
(ii) *If D is odd, then* $\operatorname{diam}\mathcal{S} \leq D$ *if and only if any two points are at distance at most $D-1$ and any two lines are at distance at most $D-1$.*

In particular:

- $\operatorname{diam}\mathcal{S} \leq 3$ *if and only if any two points are collinear and any two lines are concurrent;*
- $\operatorname{diam}\mathcal{S} \leq 4$ *if and only if for every anti-flag (p, L) some point of L is collinear with p;*
- $\operatorname{diam}\mathcal{S} \leq 6$ *if and only if for every anti-flag (p, L) some point x I L and some point y satisfy $p \in y^{\perp}$ and $x \in y^{\perp}$.*

Proof. (i) Let $p \neq p'$ be points. Choose a line L I p'. If p I L then $\mathrm{d}(p, p') = 2$; otherwise $\mathrm{d}(p, p') \leq \mathrm{d}(p, L) + 1 \leq D$. Two lines are handled in the same way, choosing a point on one of them. A flag is at distance 1. (ii) Let (p, L) be an anti-flag and choose p' I L. Then $p' \neq p$ and $\mathrm{d}(p, L) \leq \mathrm{d}(p, p') + 1 \leq D$. The converse directions of (i) and (ii) hold by parity. The three special cases are the translations of $\mathrm{d} \leq 2$, $\mathrm{d} \leq 3$ and $\mathrm{d} \leq 5$. □

2.2. **Generalized polygons.** Following Tits and [18], a *generalized n-gon* ($n \geq 3$) is an incidence structure whose incidence graph has diameter n and girth $2n$. It has *order* (s, t) if every line has $s+1$ points and every point is on $t+1$ lines; it is *thick* if $s, t \geq 2$. Elements at distance n are *opposite*. We use two consequences of the girth.

Symbol	Meaning
$\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$	incidence structure: points, lines, symmetric incidence relation
$\mathcal{S}^D = (\mathcal{L}, \mathcal{P}, \mathrm{I})$	the dual structure
$\Gamma(\mathcal{S})$	incidence graph: vertex set $\mathcal{P} \cup \mathcal{L}$, edges the flags
$\mathrm{d}(x, y)$	distance in $\Gamma(\mathcal{S})$; $\operatorname{diam}\mathcal{S} := \operatorname{diam}\Gamma(\mathcal{S})$
$\deg x$	number of elements incident with x (lines on a point; points on a line)
$\Delta(\mathcal{S})$	maximum of $\deg x$ over all elements
$x \sim y$	distinct points on a common line (collinear), or distinct lines on a common point (concurrent)
$x^\perp$	$\{x\} \cup \{y : y \sim x\}$
flag, anti-flag	incident, resp. non-incident, point–line pair
order (s, t)	every line has $s+1$ points, every point is on $t+1$ lines
opposite	at distance n in a generalized n-gon
$\mathrm{PG}(2, q)$; $(x{:}y{:}z)$, $[u{:}v{:}w]$	Desarguesian plane; points, lines; incidence $ux + vy + wz = 0$
$\mathrm{W}(q) = \mathrm{W}(3, q)$	symplectic quadrangle: points of $\mathrm{PG}(3, q)$, totally isotropic lines
$\mathrm{Q}(4, q)$, $\mathrm{Q}(6, q)$	parabolic quadrics; $\mathrm{Q}(4, q)$ is the dual of $\mathrm{W}(q)$
$\mathrm{AS}(q) = P(\mathrm{W}(q), x)$	Payne derivation of $\mathrm{W}(q)$ at x; order $(q-1, q+1)$; Ahrens–Szekeres quadrangle for q odd
$\mathrm{H}(q)$	split Cayley hexagon on $\mathrm{Q}(6, q)$, order (q, q)
$\mathcal{O}$; m-ovoid; weighted m-ovoid	ovoid; point set meeting every line in m points; weight $w\colon \mathcal{P} \to \mathbb{Z}_{\geq 0}$ with $\sum_{p \mathrm{I} L} w(p) = m$ for every line L
spread; m-cover	duals of ovoid and of (weighted) m-ovoid
$\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots)$	a filtration of a set of elements; $\lambda(x)$ the level of x
$\mathcal{S}_{\mathcal{A}}^{(m)}$, x^w	m-fold doubling of $\mathcal{S}$ along $\mathcal{A}$; the version of x indexed by the word w
$n_b(\Delta, D)$, $M_b(\Delta, D)$	largest order; bipartite Moore bound

TABLE 1. Notation.

(G1) If $\mathrm{d}(x, y) < n$, there is a unique path of length $\mathrm{d}(x, y)$ from x to y.
(G2) A generalized quadrangle ($n = 4$) of order (s, t) has $(s+1)(st+1)$ points and $(t+1)(st+1)$ lines; for an anti-flag (p, L) there is a unique point of L collinear with p. A generalized hexagon of order (s, t) has $(s+1)(s^2t^2 + st + 1)$ points and $(t+1)(s^2t^2 + st + 1)$ lines.

A generalized n-gon of order (q, q) has $2(1 + q + \cdots + q^{n-1}) = M_b(q+1, n)$ elements. By Feit and Higman [11], finite thick generalized n-gons exist only for $n \in \{3, 4, 6, 8\}$, and for $n = 8$ the number $2st$ is a square.

2.3. **The classical examples.** Let q be a prime power and $\mathbb{F} = \mathrm{GF}(q)$.

The plane $\mathrm{PG}(2, q)$ [12]. Points $(x{:}y{:}z)$ and lines $[u{:}v{:}w]$ are nonzero vectors of $\mathbb{F}^3$ up to scalars, and $(x{:}y{:}z)$ I $[u{:}v{:}w]$ if and only if $ux + vy + wz = 0$. It is a generalized 3-gon (projective plane) of order (q, q). Non-Desarguesian projective planes also exist for many q; Section 4 works in any projective plane.

The quadrangles $\mathrm{W}(q)$ *and* $\mathrm{Q}(4, q)$ [14, 3.1.1, 3.2.1]. $\mathrm{W}(q) = \mathrm{W}(3, q)$ has as points all points of $\mathrm{PG}(3, q)$ and as lines the totally isotropic lines of a symplectic polarity; $\mathrm{Q}(4, q)$ has as points and lines those of a non-singular quadric of $\mathrm{PG}(4, q)$. Both have order (q, q), and $\mathrm{Q}(4, q)$ is isomorphic to the dual of $\mathrm{W}(q)$. An *ovoid* of a generalized quadrangle is a set of points meeting every line in exactly one point; a *spread* is the dual notion. A *partial ovoid* meets every line in at most one

point. $\mathrm{Q}(4,q)$ has ovoids for every q (the elliptic quadric sections $\mathrm{Q}^-(3,q)$); $\mathrm{W}(q)$ has ovoids if and only if q is even, so $\mathrm{Q}(4,q)$ has spreads if and only if q is even [14, 3.4.1]. $\mathrm{W}(q)$ always has spreads, for instance regular spreads of $\mathrm{PG}(3,q)$.

m-ovoids and weights. An *m-ovoid* is a set of points meeting every line in exactly m points [3]. A *weighted m-ovoid* is a function $w\colon \mathcal{P} \to \mathbb{Z}_{\geq 0}$ with $\sum_{p \mathrm{I} L} w(p) = m$ for every line L; equivalently a multiset of points meeting every line in m points counted with multiplicity. Dually, a *(weighted) m-cover* is a set (multiset) of lines with exactly m members through every point. By duality, a weighted m-ovoid of $\mathrm{W}(q)$ is a weighted m-cover of $\mathrm{Q}(4,q)$.

The quadrangle $\mathrm{AS}(q)$. Every point x of $\mathrm{W}(q)$ is regular. Payne's derivation $P(\mathrm{W}(q),x)$ [13], [14, 3.1.4] has as points the points of $\mathrm{PG}(3,q)$ not in $x^\perp$, and as lines (i) the totally isotropic lines not through x, each with its point in $x^\perp$ removed, and (ii) the lines of $\mathrm{PG}(3,q)$ through x not contained in $x^\perp$ (the hyperbolic lines $\{x,y\}^{\perp\perp}$), each with x removed. It is a generalized quadrangle of order $(q-1,q+1)$, and for q odd it is isomorphic to the Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ [1], [14, 3.1.5, 3.2.6]. The lines of type (ii) form a spread of $\mathrm{AS}(q)$.

The hexagon $\mathrm{H}(q)$ [17], [18, §2.4]. Let $\mathrm{Q}(6,q)$ be the quadric $X_0X_4 + X_1X_5 + X_2X_6 = X_3^2$ of $\mathrm{PG}(6,q)$. The points of $\mathrm{H}(q)$ are all points of $\mathrm{Q}(6,q)$; the lines are the lines of $\mathrm{Q}(6,q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ satisfy Tits' equations

$$p_{12} = p_{34}, \quad p_{54} = p_{32}, \quad p_{20} = p_{35}, \quad p_{65} = p_{30}, \quad p_{01} = p_{36}, \quad p_{46} = p_{31}.$$

$\mathrm{H}(q)$ is a generalized hexagon of order (q,q). For a point x, the conditions that a point y lies on a line of $\mathrm{H}(q)$ through x are linear in y; the $q+1$ lines of $\mathrm{H}(q)$ through x are the lines through x of a totally singular plane π_x of $\mathrm{Q}(6,q)$ (the hexagon is *flat*). A *Hermitian spread* of $\mathrm{H}(q)$ is the set of lines of $\mathrm{H}(q)$ contained in a hyperplane Π for which $\Pi \cap \mathrm{Q}(6,q)$ is an elliptic quadric $\mathrm{Q}^-(5,q)$ [16], [18]; Lemma 6.1 proves the properties we use.

3. Doubling with levels

Delorme and Farhi [9] observed that a *duplicate* of an element (a new element incident with exactly the same elements) does not increase the diameter. Delorme [7, 8] refined this into a doubling of a whole set of elements, with levels, and used it to build the largest bipartite graphs then known of diameters 4 and 6. We recast it as an operation on incidence structures.[1]

3.1. **The operation.** Let $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ be an incidence structure, $m \geq 2$ an integer, and

$$\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots \supseteq \mathcal{A}_r \supseteq \mathcal{A}_{r+1} = \emptyset)$$

a filtration of a set $\mathcal{A}_0$ of elements (points and lines may be mixed). The *level* of an element x is $\lambda(x) = \max\{k : x \in \mathcal{A}_k\}$, with $\lambda(x) = -1$ if $x \notin \mathcal{A}_0$. Let $[m] = \{1, \dots, m\}$ and write $u \preceq w$ if the word u is a prefix of the word w; two words are *comparable* if one is a prefix of the other.

Definition 3.1 (Doubling)**.** The *m-fold doubling of $\mathcal{S}$ along $\mathcal{A}$* is the incidence structure $\mathcal{S}_{\mathcal{A}}^{(m)}$ whose elements are the *versions*

$$x^w, \qquad x \in \mathcal{P} \cup \mathcal{L}, \quad w \in [m]^{\lambda(x)+1},$$

where x^w is a point if x is a point and a line if x is a line, and in which

$$x^w \mathrm{I}\, y^u \iff x \mathrm{I}\, y \text{ and } w, u \text{ are comparable.}$$

An element of level -1 has the single version $x^\varnothing$, which we identify with x. For $m = 2$ we speak of *doubling*, for $m = 3$ of *tripling*.

[1] Note for co-authors: the full texts of Delorme's two 1985 papers [7, 8] have not been read by the authors of this draft. Our earlier notes attribute the doubling with levels and the bipartite constructions for $D = 4, 6$ to [7], Sections 7–8; the abstract of [8] also describes bipartite constructions. Please check which paper contains which construction; no section numbers are claimed here.

For a single level ($r = 0$, $\mathcal{A} = \mathcal{A}_0$) and $m = 2$, write $x = x^1$ and $x' = x^2$ for $x \in \mathcal{A}$. Then $\mathcal{S}_{\mathcal{A}}^{(2)}$ is obtained from $\mathcal{S}$ by adjoining, for each $x \in \mathcal{A}$, a new element x' of the same type, incident with

$$y' \text{ for every } y \in \mathcal{A} \text{ with } y \mathrel{\mathrm{I}} x, \qquad \text{and} \qquad y \text{ for every } y \notin \mathcal{A} \text{ with } y \mathrel{\mathrm{I}} x.$$

If $\mathcal{A}$ contains no flag, x' is a duplicate of x in the sense of [9]. In general x' is *not* a duplicate: inside $\mathcal{A}$ the copies are incident with copies, outside $\mathcal{A}$ with originals.

The map $x^w \mapsto x$ preserves incidence, so

$$\mathrm{d}(x^w, y^u) \geq \mathrm{d}(x, y) \qquad \text{for all versions.} \tag{1}$$

In particular two opposite elements of level -1 stay at distance n.

3.2. **Delorme's theorem.** Consider the following conditions on $\mathcal{A}$, where $D \geq 2$.

- (L0) Every element outside $\mathcal{A}_0$ is incident with at most one element of $\mathcal{A}_0$.
- (L1) For every $i \geq 0$: every element incident with an element of $\mathcal{A}_{i+1}$ lies in $\mathcal{A}_i$, and every element of $\mathcal{A}_i \setminus \mathcal{A}_{i+1}$ is incident with at most one element of $\mathcal{A}_{i+1}$.
- (C_D) For any two elements x, y (possibly $x = y$) in the same connected component of the substructure induced on $\mathcal{A}_0$, there is an element $z \notin \mathcal{A}_0$ with $\mathrm{d}(x, z) + \mathrm{d}(z, y) \leq D$.

Theorem 3.2 (Delorme [7, 8])**.** *Let $\mathcal{S}$ be an incidence structure of diameter at most D, $D \geq 2$, and let $m \geq 2$.*

(a) *If $\mathcal{A}$ satisfies* (C_D)*, then $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D.*

(b) *If $\mathcal{A}$ satisfies* (L0) *and* (L1)*, then every version x^w has $\deg x^w \leq \deg x + m - 1$, with equality only if x is incident with an element of level $\lambda(x)+1$. In particular $\Delta(\mathcal{S}_{\mathcal{A}}^{(m)}) \leq \Delta(\mathcal{S}) + m - 1$.*

(c) *The numbers of points and of lines of $\mathcal{S}_{\mathcal{A}}^{(m)}$ are*

$$|\mathcal{P}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{P}| \qquad \text{and} \qquad |\mathcal{L}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{L}|.$$

(d) *Conversely, if $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D, then* (C_D) *holds; indeed its conclusion holds for every pair $x, y \in \mathcal{A}_0$.*

We include a proof.

Proof. (c) An element of level k has m^{k+1} versions, and $m^{k+1} - 1 = (m-1)(1 + m + \cdots + m^k)$. Summing over the levels gives the formula.

(b) Let $\lambda(x) = k$ and $|w| = k + 1$. A neighbor y of x with $\lambda(y) = j \leq k$ has exactly one version incident with x^w (the one indexed by the prefix of w of length $j + 1$), and a neighbor with $\lambda(y) = j > k$ has m^{j-k} such versions. If $k = -1$, (L0) says that x has at most one neighbor in $\mathcal{A}_0$, and that neighbor has level 0, since a neighbor in $\mathcal{A}_1$ would put x in $\mathcal{A}_0$ by (L1). If $k \geq 0$, (L1) says that x has at most one neighbor in $\mathcal{A}_{k+1}$, and none in $\mathcal{A}_{k+2}$, since that would put x in $\mathcal{A}_{k+1}$. So $\deg x^w \leq \deg x - 1 + m$, with equality only when the higher neighbor exists.

(a) For an infinite word ω over $[m]$, the *sheet* Σ_ω is the set of versions x^w with $w \preceq \omega$. Every element has exactly one version in Σ_ω, and any two versions in Σ_ω have comparable words, so $x^w \mapsto x$ is an isomorphism from the substructure on Σ_ω onto $\mathcal{S}$. Two versions lie in a common sheet if and only if their words are comparable, and then they are at distance at most D.

Let x^w, y^u have incomparable words. Then both words are non-empty, so $x, y \in \mathcal{A}_0$. If x and y lie in different components of the substructure on $\mathcal{A}_0$, a shortest path $x = z_0, z_1, \ldots, z_k = y$ in $\Gamma(\mathcal{S})$, $k \leq D$, contains some $z_i \notin \mathcal{A}_0$. The element z_i has a single version, which lies in every sheet. Lift $z_0, \ldots, z_i$ into a sheet containing x^w and $z_i, \ldots, z_k$ into a sheet containing y^u; this is a path of length k from x^w to y^u. If x and y lie in the same component (this includes $x = y$), (C_D) gives $z \notin \mathcal{A}_0$ with $\mathrm{d}(x, z) + \mathrm{d}(z, y) \leq D$, and the same lifting gives a path of length at most D.

(d) Let $x, y \in \mathcal{A}_0$ (possibly $x = y$). Both have versions with non-empty words, so choose x^w and y^u with w beginning with the letter 1 and u with the letter 2. Take a path of length at most D from x^w to y^u in $\mathcal{S}_{\mathcal{A}}^{(m)}$. Consecutive versions on it have comparable words, and two comparable non-empty words begin with the same letter. So some version on the path has the empty word; it is the unique version of an element $z \notin \mathcal{A}_0$. The map $x^w \mapsto x$ preserves incidence, so the path projects to a walk from x through z to y of length at most D, whence $\mathrm{d}(x, z) + \mathrm{d}(z, y) \leq D$. □

3.3. **The hat lemma.** In a generalized polygon, condition (C_D) has a clean geometric form.

Lemma 3.3 (Hat lemma)**.** *Let $\mathcal{S}$ be a generalized n-gon of order (s, t) with $s, t \geq 1$, and let $\mathcal{A}_0$ be a set of elements. Then* (C_n) *holds if and only if every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n - 2$.*

Proof. Suppose some component B contains a path $x = z_0, z_1, \ldots, z_{n-1} = y$ of length $n - 1$. (It does if B has diameter at least $n - 1$; it does if B contains a cycle, because cycles of $\Gamma(\mathcal{S})$ have length at least $2n$.) By the girth, this is the unique path of length at most $n - 1$ from x to y, so $\mathrm{d}(x, y) = n - 1$. If $\mathrm{d}(x, z) + \mathrm{d}(z, y) \leq n$, then by parity $\mathrm{d}(x, z) + \mathrm{d}(z, y) = n - 1$, so z lies on a shortest path from x to y, that is $z = z_i \in \mathcal{A}_0$. So ($\mathrm{C}_n$) fails.

Conversely, let B be a tree of diameter at most $n-2$, let $x, y \in B$, and let P be the path from x to y in B, of length d. The component B is not all of $\Gamma(\mathcal{S})$, which contains cycles, so some element of B has a neighbor outside $\mathcal{A}_0$. Let k be the least distance in B from P to such an element u, and let $z \notin \mathcal{A}_0$ be a neighbor of u. Then $\mathrm{d}(x, z) + \mathrm{d}(z, y) \leq d + 2k + 2$. If $k = 0$ this is at most n. If $k \geq 1$, every element of B at distance less than k from P has all its neighbors in B, and every element has at least two neighbors. Walking k steps away from P from x, and k steps away from P from y (along two different neighbors if $x = y$), we obtain a path of length $d + 2k$ in the tree B. Hence $d + 2k \leq n - 2$, and again $\mathrm{d}(x, z) + \mathrm{d}(z, y) \leq n$. □

Corollary 3.4. *Let $\mathcal{S}$ be a generalized n-gon of order (s, t) with $s, t \geq 1$, let $\mathcal{A}$ be a filtration of a set $\mathcal{A}_0$ of elements, and let $m \geq 2$. Then $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most n if and only if every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n - 2$.*

Proof. By Theorem 3.2(a) and (d), the diameter is at most n if and only if (C_n) holds; apply Lemma 3.3. □

In geometric terms, the components allowed by Lemma 3.3 are:

- $n = 3$ (projective planes): single elements and flags;
- $n = 4$ (quadrangles): single elements, flags, and *stars*: a point with some of the lines on it, or a line with some of its points;
- $n = 6$ (hexagons): trees of diameter at most 4, for instance an element x, some elements incident with x, and some elements incident with those.

3.4. **Two consequences.**

Proposition 3.5 (Diameter 3 needs patches)**.** *Let $\mathcal{S}$ be a projective plane and let $\mathcal{A}$ satisfy* (L0)*,* (L1) *and* (C_3)*. Then $\mathcal{A}_0$ consists of at most one point and at most one line, and $\mathcal{A}_1 = \emptyset$.*

Proof. Let $x \neq y$ be points of $\mathcal{A}_0$. If the line xy is not in $\mathcal{A}_0$, it is incident with two elements of $\mathcal{A}_0$, against (L0). If it is in $\mathcal{A}_0$, then x, xy, y lie in one component, which is then not a single element or a flag, against Lemma 3.3. The same holds for lines. If $\mathcal{A}_1$ contained an element, its $q + 1 \geq 3$ neighbors would lie in $\mathcal{A}_0$ by (L1). □

So a pure doubling of a projective plane gains at most two elements. The constructions of Section 4 double flags and then modify a bounded number of incidences (*patches*).

Proposition 3.6 (Counting bound)**.** *Let $\mathcal{S}$ be a generalized n-gon of order (q,q) with N points and N lines, and let $\mathcal{A}_0$ satisfy* (L0) *and* (C_n)*. Let c be the number of components of the substructure on $\mathcal{A}_0$. Then*

$$q\,|\mathcal{A}_0| + 2c \le 2N,$$

with equality if and only if every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$.

Proof. By Lemma 3.3 the substructure on $\mathcal{A}_0$ is a forest with $|\mathcal{A}_0| - c$ flags. So the number of flags with exactly one element in $\mathcal{A}_0$ is $(q+1)|\mathcal{A}_0| - 2(|\mathcal{A}_0| - c)$. By (L0) it is at most $2N - |\mathcal{A}_0|$, with equality as stated. □

Since $N \sim q^{n-1}$, a doubling along $\mathcal{A}$ at $\Delta = q+2$ has $|\mathcal{A}_0| < 2N/q \sim 2q^{n-2}$; the higher levels contribute lower-order terms. Section 7 compares this with the Moore bound.

4. Diameter 3: four hats in a projective plane

By Lemma 2.1, an incidence structure without isolated elements has diameter at most 3 exactly when any two points are collinear and any two lines are concurrent. Araujo-Pardo and López [2] add three *hats* to $\mathrm{PG}(2,q)$ and obtain $n_b(q+2,3) \ge 2(q^2+q+5)$. We add a fourth hat.

4.1. **The frame.** Let Π be a projective plane of order $q \ge 3$. Fix an anti-flag (A,ℓ), and four distinct points of ℓ, labeled X_s with s in a four-element index set $S = \{0,1,a,c\}$. Put

$$T = S \setminus \{a\} = \{0,1,c\}, \qquad \lambda_s = AX_s \ (s \in S),$$

and let σ be a 3-cycle on T; for definiteness $\sigma = (0\ c\ 1)$, that is $\sigma(0) = c$, $\sigma(c) = 1$, $\sigma(1) = 0$. Let

$$\ell^* = \ell \setminus \{X_t : t \in T\} \qquad \text{and} \qquad \Lambda^* = \{\text{lines on } A\} \setminus \{\lambda_t : t \in T\}.$$

So ℓ^* consists of $q-2$ points of ℓ, among them X_a, and Λ^* of $q-2$ lines on A, among them λ_a.

Definition 4.1 (Four hats)**.** The incidence structure $\mathcal{S} = \mathcal{S}(\Pi; A, \ell, (X_s)_{s\in S}, \sigma)$ has as points the points of Π and five new points Q, H_s ($s \in S$), and as lines the lines of Π and five new lines M, K_s ($s \in S$). Its incidences are exactly the following.

(H1) The incidences of Π, except the four flags (X_s, λ_s), $s \in S$; and the three *rotated* flags $(X_t, \lambda_{\sigma(t)})$, $t \in T$:
$$X_0 \,\mathrm{I}\, \lambda_c, \qquad X_c \,\mathrm{I}\, \lambda_1, \qquad X_1 \,\mathrm{I}\, \lambda_0.$$
(H2) For $s \in S$, the hat point H_s lies on every line of Π through X_s other than ℓ.
(H3) For $s \in S$, the hat line K_s contains every point of λ_s other than A.
(H4) M contains the points of ℓ^* and the four hat points H_s.
(H5) Q lies on the lines of Λ^* and on the four hat lines K_s.
(H6) The *links*: $A \,\mathrm{I}\, K_a$, $H_a \,\mathrm{I}\, \ell$, and $H_t \,\mathrm{I}\, K_{\sigma(t)}$ for $t \in T$:
$$H_0 \,\mathrm{I}\, K_c, \qquad H_c \,\mathrm{I}\, K_1, \qquad H_1 \,\mathrm{I}\, K_0.$$

The hat pair (H_s, K_s) is a copy of the flag (X_s, λ_s): H_s lies on the lines through X_s and K_s contains the points of λ_s, with the flag "crossed" ($H_s \,\mathrm{I}\, \lambda_s$ and $X_s \,\mathrm{I}\, K_s$). The pair (Q, M) is a copy of the anti-flag (A,ℓ); see Remark 4.10.

Theorem 4.2. *Let Π be a projective plane of order $q \ge 3$. The incidence structure $\mathcal{S}$ of Definition 4.1 has q^2+q+6 points and q^2+q+6 lines, every element has degree $q+1$ or $q+2$, and $\mathcal{S}$ has diameter 3. Hence $n_b(q+2,3) \ge 2(q^2+q+6)$ for every prime power $q \ge 3$.*

The proof occupies Lemmas 4.3–4.5.

Lemma 4.3 (Degrees)**.** *In $\mathcal{S}$:*

(i) *A, every point of ℓ, Q and every H_s are on exactly $q+2$ lines;*

(ii) *a point $p \notin \ell$, $p \neq A$, of Π is on $q+2$ lines if $Ap = \lambda_s$ for some $s \in S$, and on $q+1$ lines otherwise;*

(iii) *dually, ℓ, every line on A, M and every K_s have exactly $q+2$ points, and a line $L \neq \ell$ of Π not on A has $q+2$ points if $L \cap \ell = X_s$ for some $s \in S$, and $q+1$ points otherwise.*

In particular there are $(q-1)(q-3)$ points and $(q-1)(q-3)$ lines of degree $q+1$.

Proof. We count lines on each point: the number in Π, minus removed flags, plus added flags. A: its flags in Π are untouched, and it gains K_a (H6); it is not on any other K_s by H3. $X \in \ell^*$, $X \neq X_a$: gains M. X_a: loses λ_a, gains M and K_a (H3, as $X_a \neq A$ is on λ_a). X_t, $t \in T$: loses λ_t, gains $\lambda_{\sigma(t)}$ and K_t; it is not on M. A point $p \notin \ell$, $p \neq A$: it is on exactly one line on A, namely Ap; it loses nothing and gains K_s exactly when $Ap = \lambda_s$. Q: $|\Lambda^*| + 4 = q+2$ lines. H_s: q lines of Π, M, and one link (H6): ℓ if $s = a$, $K_{\sigma(s)}$ if $s \in T$. The statement for lines is the dual count (or follows from Lemma 4.5). Finally, the points of degree $q+1$ are the $q-1$ points other than A and X on each of the $q-3$ lines AX with $X \in \ell \setminus \{X_s : s \in S\}$. □

Lemma 4.4 (Points)**.** *Any two distinct points of $\mathcal{S}$ are collinear in $\mathcal{S}$.*

Proof. The only flags of Π missing from $\mathcal{S}$ are (X_s, λ_s), $s \in S$. Let $u \neq v$ be points of $\mathcal{S}$.

(P1) u, v in Π. Let $L = uv$ in Π. If both flags survive, L works. Otherwise, say, $u = X_s$ and $L = \lambda_s$. If $v \neq A$, both u and v lie on K_s by H3. If $v = A$ and $s = a$, use K_a (H3, H6). If $v = A$ and $s \in T$, use $\lambda_{\sigma(s)}$, which contains A and, by the rotation, X_s.

(P2) Q and a point u of Π. If $u = A$, use $\lambda_a \in \Lambda^*$. If $u \in \ell$ and $u \neq X_s$ for all s, use $Au \in \Lambda^*$. If $u = X_s$, use K_s. If $u \notin \ell$, $u \neq A$, let $X = Au \cap \ell$: if $Au \in \Lambda^*$ use Au; otherwise $X = X_t$ with $t \in T$ and u, Q lie on K_t.

(P3) H_s and a point u of Π. If $u \notin \ell$ (this includes $u = A$), the line uX_s of Π is not ℓ, contains u in $\mathcal{S}$ (as $u \notin \ell$) and contains H_s by H2. If $u = X_s$, use any line through X_s other than ℓ and λ_s. If $u \in \ell^*$, $u \neq X_s$, use M. Otherwise $u = X_t$ with $t \in T \setminus \{s\}$. If $s = a$, use ℓ (H6). If $s \in T$, then $t \in \{\sigma(s), \sigma^{-1}(s)\}$ because σ is a 3-cycle. If $t = \sigma(s)$, use $K_{\sigma(s)}$, which contains $X_{\sigma(s)}$ (H3) and H_s (H6). If $t = \sigma^{-1}(s)$, use λ_s, which contains $X_{\sigma^{-1}(s)}$ (rotation) and H_s (H2).

(P4) Two new points. H_s and $H_{s'}$: M. Q and H_a: λ_a ($\lambda_a \in \Lambda^*$, and $\lambda_a \neq \ell$ is a line through X_a). *Q and H_t, $t \in T$: $K_{\sigma(t)}$.* □

Lemma 4.5 (Duality)**.** *Let Π^D be the dual plane. The dual of $\mathcal{S}(\Pi; A, \ell, (X_s), \sigma)$ is isomorphic to $\mathcal{S}(\Pi^D; \ell, A, (\lambda_s), \sigma^{-1})$, by the identity on Π and*

$$H_s \leftrightarrow K_s, \qquad Q \leftrightarrow M.$$

Consequently any two distinct lines of $\mathcal{S}$ are concurrent.

Proof. In Π^D the point ℓ is off the line A, the points λ_s lie on the line A, and the line joining the points ℓ and λ_s is X_s. Read the six rules of Definition 4.1 in Π^D with these data and with σ^{-1}. Rule H1 removes the flags (λ_s, X_s) and adds λ_t I $X_{\sigma^{-1}(t)}$, which is the rotation X_u I $\lambda_{\sigma(u)}$ of $\mathcal{S}$. Rule H2 gives a new element incident with every element of Π incident with λ_s other than A: this is K_s. Rule H3 gives a new element incident with every element incident with X_s other than ℓ: this is H_s. Rule H4 gives a new element incident with the lines on A other than λ_t ($t \in T$) and with all K_s: this is Q; likewise H5 gives M. Rule H6 gives ℓ I H_a, K_a I A and K_t I $H_{\sigma^{-1}(t)}$, which are the links of $\mathcal{S}$. Since Π^D is a projective plane of order q and σ^{-1} is a 3-cycle on T, Lemma 4.4 for $\mathcal{S}(\Pi^D; \ell, A, (\lambda_s), \sigma^{-1})$ says that any two lines of $\mathcal{S}$ are concurrent. □

Proof of Theorem 4.2. The counts are Lemma 4.3; no element is isolated. By Lemmas 4.4, 4.5 and 2.1, $\operatorname{diam} \mathcal{S} \leq 3$, and $\operatorname{diam} \mathcal{S} = 3$ because $A \not\mathrel{I} \ell$. □

4.2. **Coordinates and a regular completion.** Now let $\Pi = \mathrm{PG}(2,q)$ with $q \geq 4$. Take

$$\ell = [1{:}0{:}0], \quad A = (1{:}0{:}0), \quad P = (0{:}1{:}0), \quad \lambda_\infty = [0{:}0{:}1], \qquad X_t = (0{:}t{:}1), \quad \lambda_t = [0{:}1{:}{-}t] \quad (t \in \mathbb{F}),$$

so that $\ell = \{P\} \cup \{X_t\}$, the lines on A are λ_∞ and the λ_t, $\lambda_t \cap \ell = X_t$ and $\lambda_\infty \cap \ell = P$. Let $a, c \in \mathbb{F} \setminus \{0,1\}$ be distinct, and use the labels $S = \{0,1,a,c\} \subseteq \mathbb{F}$ for the points X_0, X_1, X_a, X_c. Then $\ell^* = \{P\} \cup \{X_t : t \notin T\}$ and $\Lambda^* = \{\lambda_\infty\} \cup \{\lambda_t : t \notin T\}$, and the rules read:

- H_s ($s \in S$) lies on $[w{:}1{:}{-}s]$ for every $w \in \mathbb{F}$ (these are the lines on X_s other than ℓ, with λ_s at $w = 0$);
- K_s ($s \in S$) contains $(x{:}s{:}1)$ for every $x \in \mathbb{F}$ (the points of λ_s other than A, with X_s at $x = 0$);
- M contains P, X_t ($t \notin T$) and the H_s; Q lies on λ_∞, λ_t ($t \notin T$) and the K_s;
- rotation and links as in H1 and H6.

By Lemma 4.3 the elements of degree $q+1$ are the points $(x{:}y{:}1)$ ($x \neq 0$, $y \notin S$) and $(x{:}1{:}0)$ ($x \neq 0$), and the lines $[u{:}1{:}w]$ ($u \neq 0$, $-w \notin S$) and $[1{:}0{:}w]$ ($w \neq 0$).

Lemma 4.6 (A correlation)**.** *The map* $\delta\colon (x{:}y{:}z) \mapsto [x{:}z{:}{-}y]$, $[u{:}v{:}w] \mapsto (u{:}w{:}{-}v)$ *is a correlation of* $\mathrm{PG}(2,q)$ *with*

$$\delta(A) = \ell, \quad \delta(X_t) = \lambda_t, \quad \delta(P) = \lambda_\infty,$$

and a point p lies on $\delta(p)$ if and only if $p \in \ell$. It maps the points of degree $q+1$ bijectively onto the lines of degree $q+1$.

Proof. $ux + wz + (-v)(-y) = ux + vy + wz$, so δ preserves incidence; it is clearly bijective. $(x{:}y{:}z)$ I $[x{:}z{:}{-}y]$ iff $x^2 + yz - zy = x^2 = 0$. Finally $\delta(x{:}y{:}1) = [x{:}1{:}{-}y]$ with $x \neq 0$, $y \notin S$, and $\delta(x{:}1{:}0) = [x{:}0{:}{-}1] = [1{:}0{:}{-}1/x]$. □

Theorem 4.7. *Let $q \geq 4$ be a prime power and a, c distinct elements of $\mathbb{F} \setminus \{0,1\}$. Adjoining to $\mathcal{S}(\mathrm{PG}(2,q); A, \ell, (X_s), \sigma)$ the $(q-1)(q-3)$ flags $(p, \delta(p))$, p a point of degree $q+1$, gives a $(q+2)$-regular incidence structure with q^2+q+6 points, q^2+q+6 lines and diameter 3.*

Proof. By Lemma 4.6 the new flags are not flags of $\mathcal{S}$ (the points of degree $q+1$ are off ℓ) and they match the elements of degree $q+1$ in pairs. Adding flags cannot increase distances, and $A \not\mid \ell$ still. □

Remark 4.8 (Arbitrary planes)*.* In any projective plane of order $q \geq 7$ a regular completion also exists. Each of the $N = (q-1)(q-3)$ points of degree $q+1$ is incident with at most $q+1$ of the N lines of degree $q+1$, and $N - (q+1) \geq N/2$ for $q \geq 7$, so Hall's theorem gives a perfect matching of non-incident pairs.

Remark 4.9 (The three-hat structure of [2])*.* In the same coordinates, the structure of Araujo-Pardo and López uses the hats $S_{AL} = \{0,1,a\}$, no rotation, and links A I K_a, H_a I ℓ, H_0 I K_1, H_1 I K_0; it removes only the flag (X_a, λ_a), and M contains P and X_t for $t \notin \{0,1\}$, Q lies on λ_∞ and λ_t for $t \notin \{0,1\}$. (In their affine coordinates $(x,y) = (x{:}y{:}1)$, $[m,b] = [m{:}{-}1{:}b]$, $L_c = [1{:}0{:}{-}c]$, $L = [0{:}0{:}1]$, $P_m = (1{:}m{:}0)$; so $L_0 = \ell$, $P_0 = A$, $(0,s) = X_s$, $[0,s] = \lambda_s$ and $L = \lambda_\infty$.) It has $2(q^2+q+5)$ elements and diameter 3 for $q \geq 3$ [2]; the proof is a case analysis like that of Lemma 4.4. Its elements of degree $q+1$ are the points (x,y), $x \neq 0$, $y \notin S_{AL}$, and P_i, $i \neq 0$, and the lines $[x,y]$, $x \neq 0$, $y \notin S_{AL}$, and L_i, $i \neq 0$. The completing matching printed in [2] is $(x,y) \sim [x,y]$ together with $P_i \sim L_i$ for $i \in \mathbb{F} \setminus \{0,1,a\}$; with this index set the four elements P_1, P_a, L_1, L_a keep degree $q+1$. Taking $i \in \mathbb{F}^*$ instead (the pairs $P_i \notin L_i$ are always non-incident) gives a $(q+2)$-regular structure, as intended. This affects neither the maximum degree nor the diameter.

Remark 4.10 (Hats as doubled flags)*.* Let $\mathcal{A} = \{\ell, A\} \cup \{X_s, \lambda_s : s \in S_{AL}\}$ in $\mathrm{PG}(2,q)$ and double it ($m = 2$), except that each flag (X_s, λ_s) is *crossed*: instead of X_s' I λ_s' put X_s' I λ_s and X_s I λ_s'. With $M = \ell'$, $Q = A'$, $H_s = X_s'$, $K_s = \lambda_s'$, one checks from the rules that the structure of [2] is this crossed doubling plus the six incidences H_1 I K_0, H_0 I K_1, H_a I ℓ, A I K_a, X_a I M, Q I λ_a, minus

the flag (X_a, λ_a). By Proposition 3.5 some patching is unavoidable: the component of $\mathcal{A}$ contains the path ℓ, X_s, λ_s, A of length $3 > n-2 = 1$. For $q = 5$ a direct check shows that in the crossed doubling without patches the pairs at distance greater than 3 are exactly (H_s, X_t), (K_s, λ_t) $(s \neq t)$, (H_s, Q) and (K_s, M), all of them copies of pairs joined inside $\mathcal{A}$ by a path of length $2 = n-1$ (such as X_s, ℓ, X_t or X_s, λ_s, A), as in the proof of Lemma 3.3; the patches repair exactly these. Heuristically, the fourth hat needs the rotation because M now misses three points of ℓ, and each hat point H_t, $t \in T$, must reach the two of them other than X_t with only two spare incidences.

Remark 4.11 (Scope). All proofs above use only the axioms of a projective plane, so Theorem 4.2 holds in non-Desarguesian planes as well. For $\Delta = q+2 \leq 16$ the theorem gives 192, 276, 376 at $\Delta = 11, 13, 15$ $(q = 9, 11, 13)$, two more than [2]; at $\Delta = 7$ $(q = 5)$ it gives 72, below the known value 80.

5. Diameter 4: generalized quadrangles

By Lemma 2.1, an incidence structure without isolated elements has diameter at most 4 exactly when, for every anti-flag (p, L), some point of L is collinear with p. Every generalized quadrangle has this property, with a unique such point.

5.1. **Doubling in quadrangles.** By Lemma 3.3, in a generalized quadrangle a set $\mathcal{A}_0$ satisfies (C_4) exactly when its components are single elements, flags or stars. The following is Delorme's construction [7, 8] in this language.

Example 5.1 (Delorme)**.** Let Σ be a spread of $\mathrm{W}(q)$, let $v \in \Sigma$, and let $\mathcal{P}(v)$ be the set of points of v. Put

$$\mathcal{A}_1 = \{v\}, \qquad \mathcal{A}_0 = \Sigma \cup \mathcal{P}(v).$$

(L1): the elements incident with v are its points; a point of v is incident with v only, a line of $\Sigma \setminus \{v\}$ with nothing in $\mathcal{A}_1$. (L0): a point off v is on exactly one line of Σ and on no other line of $\mathcal{A}_0$; a line not in Σ meets v in at most one point and contains no other element of $\mathcal{A}_0$. The components are the star $\{v\} \cup \mathcal{P}(v)$ and the single lines of $\Sigma \setminus \{v\}$. By Theorem 3.2 with $m = 2$ we get maximum degree $q+2$, diameter 4 and

$$2(q+1)(q^2+1) + |\mathcal{A}_0| + 2|\mathcal{A}_1| = 2(q+1)(q^2+1) + q^2 + q + 4 = 2q^3 + 3q^2 + 3q + 6$$

elements. These are the previous table values 346, 1734, 3064, 4946 at $\Delta = 7, 11, 13, 15$. Dually: an ovoid $\mathcal{O}$ of $\mathrm{Q}(4, q)$, a point $o \in \mathcal{O}$, $\mathcal{A}_1 = \{o\}$, $\mathcal{A}_0 = \mathcal{O} \cup \{\text{lines on } o\}$.

Remark 5.2 (Ovoids and stars). More generally, let $\mathcal{O}$ be an ovoid of $\mathrm{Q}(4, q)$ and $\mathcal{M}$ a set of lines such that every point outside $\mathcal{O}$ is on at most one line of $\mathcal{M}$. Then $\mathcal{A}_0 = \mathcal{O} \cup \mathcal{M}$ (one level) satisfies (L0) and (C_4): its components are stars centered at points of $\mathcal{O}$. This gives maximum degree $q+2$, diameter 4 and $2(q+1)(q^2+1) + q^2 + 1 + |\mathcal{M}|$ elements. If some $o \in \mathcal{O}$ is on $k \geq 2$ lines of $\mathcal{M}$, then $|\mathcal{M}| \leq k + (q+1-k)q \leq q^2 - q + 2$. Indeed, a line $L \in \mathcal{M}$ not on o has a unique point y collinear with o; $y \notin \mathcal{O}$, the line oy is not in $\mathcal{M}$ (else y would be on two lines of $\mathcal{M}$), and distinct L give distinct y. If every point of $\mathcal{O}$ is on at most one line of $\mathcal{M}$, then $\mathcal{M}$ is a partial spread, so $|\mathcal{M}| \leq q^2 - q + 1$ for q odd (Tallini [15][2]; for large partial ovoids see also [4]).

5.2. **Clones and shadow lines.** The next construction is not a doubling: it adjoins several copies of points and copies of lines that are attached to copies of points.

Definition 5.3 (Clones and shadows)**.** Let $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ be a generalized quadrangle of order (s, t). Let $w \colon \mathcal{P} \to \mathbb{Z}_{\geq 0}$ be a weight and $\mathcal{M}$ a finite multiset of lines. For each entry μ of $\mathcal{M}$, with underlying line L_μ, choose for every point p I L_μ a *stand-in* $\sigma_\mu(p) \in \{p, p^1, \ldots, p^{w(p)}\}$. The structure $\mathcal{S}[w, \mathcal{M}, \sigma]$ has

[2]Note for co-authors: the page range of this reference (Mitt. Math. Sem. Giessen 201 (1991) 141–147) is quoted from memory and has not been verified.

- points: $\mathcal{P}$ and the *clones* $p^1, \dots, p^{w(p)}$ of each $p \in \mathcal{P}$;
- lines: $\mathcal{L}$ and one *shadow line* L^μ for each entry μ of $\mathcal{M}$;
- incidences: those of $\mathcal{S}$; $p^i \mathrel{\mathrm{I}} L$ whenever $p \mathrel{\mathrm{I}} L \in \mathcal{L}$; and $\sigma_\mu(p) \mathrel{\mathrm{I}} L^\mu$ for $p \mathrel{\mathrm{I}} L_\mu$.

So a clone is incident with exactly the old lines through its point, and a shadow line has exactly one point "over" each point of its line. Write π for the map sending a clone or a shadow line to the underlying point or line (π is the identity on $\mathcal{P} \cup \mathcal{L}$).

Theorem 5.4. *Let $\mathcal{S}$ be a generalized quadrangle of order (s,t), $s,t \geq 1$, and let $\mathcal{S}' = \mathcal{S}[w, \mathcal{M}, \sigma]$.*

(a) *$\mathcal{S}'$ has diameter 4.*

(b) *$\Delta(\mathcal{S}') \leq \Delta$ if and only if $\Delta \geq \max(s,t)+1$, $\sum_{p \mathrm{I} L} w(p) \leq \Delta - s - 1$ for every line L of $\mathcal{S}$, and every point of $\mathcal{S}'$ (old point or clone) is incident with at most $\Delta - t - 1$ shadow lines.*

(c) *$\mathcal{S}'$ has $(s+1)(st+1) + \sum_p w(p)$ points and $(t+1)(st+1) + |\mathcal{M}|$ lines.*

Proof. (b) and (c) are counts: an old line L has $s+1+\sum_{p \mathrm{I} L} w(p)$ points, a shadow line $s+1$ points, and an old point or clone is on $t+1$ old lines plus the shadow lines it serves.

(a) We use Lemma 2.1(i). Let (x, Y) be an anti-flag of $\mathcal{S}'$. Every point x of $\mathcal{S}'$ is incident with every old line through $\pi(x)$. If Y is an old line, then $\pi(x) \not\mathrel{\mathrm{I}} Y$; let r be the point of Y collinear with $\pi(x)$ in $\mathcal{S}$, on the line N. Then $x \mathrel{\mathrm{I}} N \mathrel{\mathrm{I}} r \mathrel{\mathrm{I}} Y$. If $Y = L^\mu$ and $\pi(x) \not\mathrel{\mathrm{I}} L_\mu$, let $r \mathrel{\mathrm{I}} L_\mu$ be collinear with $\pi(x)$ on the line N. Then $x \mathrel{\mathrm{I}} N \mathrel{\mathrm{I}} \sigma_\mu(r) \mathrel{\mathrm{I}} L^\mu$. If $Y = L^\mu$ and $\pi(x) \mathrel{\mathrm{I}} L_\mu$, pick $r \mathrel{\mathrm{I}} L_\mu$ with $r \neq \pi(x)$. Then $x \mathrel{\mathrm{I}} L_\mu \mathrel{\mathrm{I}} \sigma_\mu(r) \mathrel{\mathrm{I}} L^\mu$. In each case some point of Y is collinear with x, so $\operatorname{diam} \mathcal{S}' \leq 4$. Two non-collinear points p, p' of $\mathcal{S}$ remain non-collinear: they share no old line, and a shadow line L^μ contains only points over points of L_μ, and an old point is over itself only. So $\operatorname{diam} \mathcal{S}' = 4$. □

If all stand-ins are the old points, a shadow line is a duplicate of its line except that it misses the clones; this is the setting of the following corollary.

Corollary 5.5. *Let $\mathcal{S}$ be a generalized quadrangle of order (s,t) and $\Delta \geq \max(s,t)+1$. Suppose $\mathcal{S}$ has a weighted $(\Delta - s - 1)$-ovoid w and a weighted $(\Delta - t - 1)$-cover $\mathcal{M}$. Then $\mathcal{S}[w, \mathcal{M}, \mathrm{id}]$ has diameter 4, maximum degree Δ and $2\Delta(st+1)$ elements. Adjoining further flags between clones and shadow lines makes it Δ-regular, with the same number of elements and diameter 4.*

Proof. Double counting flags gives $\sum_p w(p) = (\Delta - s - 1)(st+1)$ and $|\mathcal{M}| = (\Delta - t - 1)(st+1)$, and the total is $(s+t+2)(st+1) + (\Delta - s - 1)(st+1) + (\Delta - t - 1)(st+1) = 2\Delta(st+1)$. Old points and old lines have degree exactly Δ. Each clone has degree $t+1$ and lacks $\Delta - t - 1$; each shadow line has degree $s+1$ and lacks $\Delta - s - 1$; and $\sum_p w(p)\,(\Delta - t - 1) = |\mathcal{M}|\,(\Delta - s - 1)$. Order the clones $c_0, c_1, \dots$ and the shadow lines $y_0, \dots, y_{b-1}$, $b = |\mathcal{M}|$, and join c_i to y_j for $j \equiv i(\Delta - t - 1) + k \pmod b$, $0 \leq k < \Delta - t - 1$. Since $\Delta - t - 1 \leq b$ these are distinct, and every shadow line receives exactly $\Delta - s - 1$ of them. Adding flags cannot increase distances, and two non-collinear old points still have no common line. □

5.3. **Two elliptic quadrics in** $\mathrm{AS}(q)$**.** Let q be odd. Take $\mathrm{W}(q)$ with the alternating form $\beta(u,v) = u_0v_1 - u_1v_0 + u_2v_3 - u_3v_2$ and the point $x = (1{:}0{:}0{:}0)$. Then $x^\perp$ is the plane $X_1 = 0$, and the points of $\mathrm{AS}(q) = P(\mathrm{W}(q), x)$ are the q^3 points $(a{:}1{:}w_1{:}w_2)$, which we write (a, w) with $a \in \mathbb{F}$, $w = (w_1, w_2) \in \mathbb{F}^2$. The lines of $\mathrm{AS}(q)$ are:

- the *fibers* $\{(a_0 + \mu, w_0) : \mu \in \mathbb{F}\}$: the hyperbolic lines xy, $y \notin x^\perp$, with x removed;
- the sets $\{(a_0 + \lambda \det(w_0, d),\ w_0 + \lambda d) : \lambda \in \mathbb{F}\}$ with $d \neq 0$: the totally isotropic line through (a_0, w_0) and $(\det(w_0, d){:}0{:}d)$, with its point in $x^\perp$ removed.

Fix a non-square $\varepsilon \in \mathbb{F}$ and let $\mathrm{N}(w) = w_1^2 - \varepsilon w_2^2$, an anisotropic form. For $\kappa \in \mathbb{F}^*$ let

$$\mathcal{E}_\kappa\colon\ X_0X_1 = \kappa\,\mathrm{N}(X_2, X_3).$$

This is an elliptic quadric of $\mathrm{PG}(3,q)$ through x, with tangent plane $x^\perp$ at x. So $\mathcal{E}_\kappa \cap x^\perp = \{x\}$ and $\mathcal{E}_\kappa \setminus \{x\} = \{(\kappa \mathrm{N}(w), w) : w \in \mathbb{F}^2\}$ is a set of q^2 points of $\mathrm{AS}(q)$.

Theorem 5.6. *Let q be odd, ε a non-square and $\kappa_1\kappa_2 = 1/(4\varepsilon)$. Then every line of* $\mathrm{AS}(q)$ *meets $\mathcal{E}_{\kappa_1} \setminus \{x\}$ and $\mathcal{E}_{\kappa_2} \setminus \{x\}$ in exactly two points in total, counted with multiplicity. The two quadrics meet only in x and $e = (0,0)$. Hence $w = \mathbf{1}_{\mathcal{E}_{\kappa_1}\setminus x} + \mathbf{1}_{\mathcal{E}_{\kappa_2}\setminus x}$ is a weighted 2-ovoid of* $\mathrm{AS}(q)$ *with $w(e) = 2$.*

Proof. Fibers. A fiber lies on a line of $\mathrm{PG}(3,q)$ through x not in the tangent plane $x^\perp$, so it meets each $\mathcal{E}_\kappa$ in one point besides x (in coordinates, at $\mu = \kappa\mathrm{N}(w_0) - a_0$).

The other lines. Write $\rho = \det(w_0, d)$ and $B = \mathrm{N}(w_0 + d) - \mathrm{N}(w_0) - \mathrm{N}(d)$. The point with parameter λ lies on $\mathcal{E}_\kappa$ if and only if

$$\kappa\mathrm{N}(d)\lambda^2 + (\kappa B - \rho)\lambda + \kappa\mathrm{N}(w_0) - a_0 = 0.$$

As $\mathrm{N}(d) \neq 0$ this is a quadratic with $1+\chi(\mathrm{disc}_\kappa)$ roots, where χ is the quadratic character ($\chi(0) = 0$) and disc_κ the discriminant. With $w_0 = (b_0, c_0)$, $d = (b_1, c_1)$ one has $B = 2(b_0b_1 - \varepsilon c_0c_1)$ and

$$B^2 - 4\mathrm{N}(w_0)\mathrm{N}(d) = 4\varepsilon(b_0c_1 - c_0b_1)^2 = 4\varepsilon\rho^2,$$

hence

$$\mathrm{disc}_\kappa = (\kappa B - \rho)^2 - 4\kappa\mathrm{N}(d)(\kappa\mathrm{N}(w_0) - a_0) = 4\kappa\mathrm{N}(d)\,(a_0 - a^*_\kappa), \qquad a^*_\kappa = -\frac{4\varepsilon\kappa\rho^2 - 2B\rho + \rho^2/\kappa}{4\mathrm{N}(d)}.$$

Now $a^*_{\kappa_1} - a^*_{\kappa_2} = -\frac{\rho^2}{4\mathrm{N}(d)}(\kappa_1 - \kappa_2)\big(4\varepsilon - \frac{1}{\kappa_1\kappa_2}\big) = 0$; call the common value a^*. The line meets the two quadrics in

$$2 + \chi\big(\mathrm{N}(d)(a_0 - a^*)\big)\big(\chi(\kappa_1) + \chi(\kappa_2)\big) = 2$$

points, because $\chi(\kappa_1)\chi(\kappa_2) = \chi(\varepsilon) = -1$. (A double root, $a_0 = a^*$, gives one point on each quadric.)

Intersection. $\kappa_1 \neq \kappa_2$, since $\kappa_1^2 = 1/(4\varepsilon)$ is impossible. So $\kappa_1\mathrm{N}(w) = \kappa_2\mathrm{N}(w)$ forces $w = 0$. □

Corollary 5.7. *For every odd prime power q,* $\mathrm{AS}(q)[w, \emptyset]$ *with w as in Theorem 5.6 is $(q+2)$-regular of diameter 4 with $2q^2(q+2)$ elements. Hence $n_b(q+2, 4) \geq 2q^2(q+2)$.*

Proof. Theorem 5.4 with $(s,t) = (q-1, q+1)$ and $\mathcal{M} = \emptyset$: every line gains exactly two points, every clone is on the $q+2$ lines of its point, and there are $q^3 + (q+2)q^2 + 2q^2$ elements. □

At $q = 9, 11, 13$ this gives 1782, 3146, 5070. Since the structure is regular, no choice of clones and shadows on $\mathrm{AS}(q)$ does better at $\Delta = q + 2$. Weights are needed: for $q = 3, 5$, $\mathrm{AS}(q)$ has no 2-ovoid without repeated points.

5.4. **A Singer 2-ovoid of** $\mathrm{W}(q)$**.** Let q be odd, $\mathbb{F} = \mathrm{GF}(q) \subset K = \mathrm{GF}(q^2) \subset E = \mathrm{GF}(q^4)$, and g a primitive element of E. Model $\mathrm{PG}(3,q)$ as $E^*/\mathbb{F}^*$; the point $\langle g^i\rangle$ has index i modulo $(q+1)(q^2+1)$. Write $\bar{y} = y^{q^2}$, $k' = k^q$ for $k \in K$, $\mathrm{N}(y) = y\bar{y} \in K$ and $\mathrm{Tr} = \mathrm{Tr}_{K/\mathbb{F}}$. Let $d = g^{(q^2+1)/2}$, so $\bar{d} = -d$ and $h = d^2 = g^{q^2+1}$ generates K^*. The form

$$b(y,z) = \mathrm{Tr}_{E/\mathbb{F}}(d\,y\bar{z}) = \mathrm{Tr}\big(d(y\bar{z} - \bar{y}z)\big)$$

is alternating and non-degenerate; let $\mathrm{W}(q)$ be its symplectic quadrangle. Multiplication by g^{q+1} scales b by an element of $\mathbb{F}^*$, so it preserves $\mathrm{W}(q)$. Its orbits on points are the classes

$$C_a = \{\langle g^i\rangle : i \equiv a \pmod{q+1}\}, \qquad |C_a| = q^2 + 1,$$

and $\langle y\rangle \in C_a$ if and only if $\mathrm{N}(y) \in h^a\mathbb{F}^*$. These are Ebert's elliptic quadrics partitioning $\mathrm{PG}(3,q)$ [10].

Proposition 5.8. *Let q be odd.*

(a) *Every line of* $\mathrm{PG}(3,q)$ *meets each class C_a in at most two points.*
(b) *Every totally isotropic line of* $\mathrm{W}(q)$ *meets $C_a \cup C_{q-a}$ in exactly two points.*

So the $(q+1)/2$ sets $C_a \cup C_{q-a}$, $0 \leq a \leq (q-1)/2$, are 2-ovoids of $\mathrm{W}(q)$ *partitioning its points; dually, they are 2-covers of* $\mathrm{Q}(4,q)$.

Proof. Let $L = \langle y, z\rangle_{\mathbb{F}}$ and $u = sy + tz$. Then $\mathrm{N}(u) = As^2 + \beta st + Ct^2$ with $A = \mathrm{N}(y)$, $C = \mathrm{N}(z)$, $\beta = y\bar{z} + \bar{y}z$, all in K. Put $\theta = y\bar{z} - \bar{y}z$; as $\bar{\theta} = -\theta$, $\theta = dk_0$ with $k_0 \in K$, and

$$\mathcal{D} := \beta^2 - 4AC = \theta^2 = hk_0^2, \qquad b(y,z) = \mathrm{Tr}(hk_0).$$

Fix $\omega \in K^*$ with $\omega' = -\omega$. For $\mu \in K^*$ put $\nu = \omega/\mu$; then $\mathrm{Tr}(\nu k) = 0$ if and only if $k \in \mu\mathbb{F}$. So $\langle u\rangle$ lies in the class $\{\langle u\rangle : \mathrm{N}(u) \in \mu\mathbb{F}^*\}$ if and only if $(s{:}t)$ is a zero of

$$f_\nu(s,t) = \mathrm{Tr}(\nu A)s^2 + \mathrm{Tr}(\nu\beta)st + \mathrm{Tr}(\nu C)t^2.$$

(a) Suppose $f_\nu \equiv 0$. Then $A, \beta, C \in \mu\mathbb{F}$. If $\mathcal{D} \neq 0$, then $h = \mathcal{D}/k_0^2 \in \mu^2\mathbb{F}^*/k_0^2$; but $\mathbb{F}^* \subseteq (K^*)^2$, as $q{+}1$ is even, so h would be a square in K^*, which it is not. If $\mathcal{D} = 0$, then $\mathrm{N}(u) = \mu(\alpha_0 s^2 + \beta_0 st + \gamma_0 t^2)$ with $\beta_0^2 = 4\alpha_0\gamma_0$ and $\alpha_0 \neq 0$ (as $A \neq 0$), which vanishes at $(s{:}t) = (-\beta_0 : 2\alpha_0)$, impossible for $u \neq 0$. So $f_\nu \not\equiv 0$, and it has $1 + \chi(\mathrm{disc}(\nu)) \leq 2$ zeros, $\mathrm{disc}(\nu) = \mathrm{Tr}(\nu\beta)^2 - 4\mathrm{Tr}(\nu A)\mathrm{Tr}(\nu C)$.

(b) Expanding, $\mathrm{disc}(\nu) = \mathrm{Tr}(\nu^2\mathcal{D}) + \nu\nu' P$ with $P = 2\beta\beta' - 4AC' - 4A'C \in \mathbb{F}$. If L is totally isotropic then $hk_0 \in \omega\mathbb{F}$, so $e := h\mathcal{D} = (hk_0)^2 \in \mathbb{F}$, and

$$\mathrm{Tr}(h^2\nu'^2\mathcal{D}) = e(h\nu'^2 + h'\nu^2) = hh'\,\mathrm{Tr}(\nu^2\mathcal{D}).$$

Hence $\mathrm{disc}(h\nu') = \mathrm{N}_{K/\mathbb{F}}(h)\,\mathrm{disc}(\nu)$. Since h generates K^*, $\mathrm{N}_{K/\mathbb{F}}(h)$ generates $\mathbb{F}^*$ and is a non-square, so $\chi(\mathrm{disc}(h\nu')) = -\chi(\mathrm{disc}(\nu))$. The class belonging to $h\nu'$ is $\omega/(h\nu') \equiv \mu'/h \equiv (h\mu)^{-1}$ modulo $\mathbb{F}^*$ (using $\mu^{q+1} \in \mathbb{F}^*$); in indices, $a \mapsto q - a$. Scaling ν by $\mathbb{F}^*$ multiplies the discriminant by a square. So L meets $C_a \cup C_{q-a}$ in $(1 + \chi(\mathrm{disc}(\nu))) + (1 - \chi(\mathrm{disc}(\nu))) = 2$ points. The pairing $a \leftrightarrow q - a$ has no fixed point since $q + 1$ is even. □

Remark 5.9. For q odd, $\mathrm{W}(q)$ is known to have m-ovoids for every even m (in particular 2-ovoids) and $(q+1)/2$-ovoids [3]; see also [6] for m-ovoids of $\mathrm{W}(q)$. We have not checked whether the 2-ovoids $C_a \cup C_{q-a}$, or the partition they form, coincide with those constructions; the proof above is self-contained.[3] The proof only uses that d^2 is a non-square in K: replacing d by any δ_0 with $\bar{\delta}_0 = -\delta_0$ gives a symplectic form for which $C_a \cup C_b$ is a 2-ovoid when $h^{a+b}\delta_0^2 \in \mathbb{F}^*$.

5.5. **All degrees.**

Corollary 5.10. *Let q be a prime power and $\Delta \geq q + 1$. There is a Δ-regular incidence structure of diameter 4 with*

$$2\Delta(q^2+1) \text{ elements if } q \text{ is even or } \Delta - q - 1 \text{ is even}, \qquad 2\Delta q^2 \text{ elements otherwise.}$$

In particular $n_b(\Delta, 4) \geq 2q^2\Delta$ for every prime power $q \leq \Delta - 1$.

Proof. Apply Corollary 5.5. Let $j = \Delta - q - 1$. If q is even, $\mathrm{Q}(4,q)$ has an ovoid $\mathcal{O}$ and a spread Σ; take $w = j\mathbf{1}_{\mathcal{O}}$ and $\mathcal{M} = j\Sigma$. If q is odd and j is even, take $w = j\mathbf{1}_{\mathcal{O}}$ and $\mathcal{M}$ equal to $j/2$ copies of a 2-cover of $\mathrm{Q}(4,q)$ from Proposition 5.8. If q is odd and j is odd, then $\Delta \geq q + 2$; in $\mathrm{AS}(q)$, of order $(q-1, q+1)$, take $(\Delta - q)/2$ copies of the weighted 2-ovoid of Theorem 5.6 and $\Delta - q - 2$ copies of the spread of fibers. □

At $q = 13$, $\Delta = 16$ this gives $n_b(16, 4) \geq 5440$.

5.6. **Limits of the method.**

Proposition 5.11. *Let q be odd, $\mathcal{O}$ an ovoid of $\mathrm{Q}(4,q)$, $w = \mathbf{1}_{\mathcal{O}}$, and let $\mathcal{S}' = \mathrm{Q}(4,q)[w, \mathcal{M}, \sigma]$ have maximum degree $q + 2$. Then $|\mathcal{M}| \leq q^2 - q + 2$.*

[3]Note for co-authors: we have confirmed from the abstract of [3] only the existence of m-ovoids of $\mathrm{W}(q)$ for even m and of $(q+1)/2$-ovoids, q odd. Whether partitions into 2-ovoids occur there, and whether [6] has a Singer-group description, still has to be checked.

Proof. By Theorem 5.4(b) every old point and every clone serves at most one shadow line. So a point outside $\mathcal{O}$ is on at most one entry of $\mathcal{M}$, and a point $o \in \mathcal{O}$ on at most two (one served by o, one by its clone). In particular $\mathcal{M}$ has no repeated line. If some $o \in \mathcal{O}$ is on two lines $L_1, L_2 \in \mathcal{M}$, then every other $L \in \mathcal{M}$ misses o, and its unique point y collinear with o is not in $\mathcal{O}$ (points of an ovoid are pairwise non-collinear). The line oy is not in $\mathcal{M}$, since y is on at most one line of $\mathcal{M}$, so oy is one of the $q-1$ lines on o other than L_1, L_2, and y is one of its q points other than o. Distinct L give distinct y, so $|\mathcal{M}| \le 2 + q(q-1)$. Otherwise $\mathcal{M}$ is a partial spread of $\mathrm{Q}(4,q)$ and $|\mathcal{M}| \le q^2 - q + 1$ (Tallini [15]). □

So these structures have at most $2(q+1)(q^2+1) + (q^2+1) + (q^2-q+2) = 2q^2(q+2) + q + 5$ elements; to beat Corollary 5.7 one needs $|\mathcal{M}| \ge q^2 - 2q - 2$. For $q = 5$ there is a set $\mathcal{M}$ of 21 lines with suitable stand-ins (an explicit example, checked by computer), giving $n_b(7,4) \ge 359$; we do not know whether $q^2 - 2q - 2$ lines are possible for $q \ge 9$. In particular, $q^2 + 1$ shadow lines (which would give $2(q^2+1)(q+2)$ elements) are impossible for odd q.

Remark 5.12 (Weighted 3-ovoids). Weighted 3-ovoids of $\mathrm{W}(q)$, that is weighted 3-covers of $\mathrm{Q}(4,q)$, exist at least for $q = 3, 5, 7$ (found by computer; data in `research/p4/d4/sols/w_q3_j3.json`, `research/p4/d4/sols/w_q5_j3.json`, `research/p4/d4/sols/w3_q7.json`). For $q = 5$ there is even a 3-ovoid, and the $(q+1)/2$-ovoids of [3] are 3-ovoids exactly when $q = 5$. Together with 2-covers they would remove the case distinction in Corollary 5.10 for $\Delta \ge q + 4$.

Question 5.13. Does $\mathrm{W}(q)$ have a weighted 3-ovoid for every odd q? Does $\mathrm{AS}(q)$ have a weighted 3-ovoid for every odd q (known for $q = 5, 7$ by computer search; logs `research/gq4x/runs/ilp_q5_m3_exact_mult3.log` and `research/gq4x/runs/ilp_q7_m3_exact_mult3.log`)?

6. Diameter 6: the split Cayley hexagon

Throughout this section q is odd, $\mathbb{F} = \mathrm{GF}(q)$, and $\mathrm{H}(q)$ is the split Cayley hexagon on the quadric $\mathrm{Q}(6,q)$ of equation $X_0X_4 + X_1X_5 + X_2X_6 = X_3^2$, as in Section 2. Distances are taken in the incidence graph of $\mathrm{H}(q)$; two lines are opposite when they are at distance 6, and a point is at distance 1, 3 or 5 from a line.

6.1. **The frame.** Let $v = \langle e_1, e_6 \rangle$. It is a line of $\mathrm{H}(q)$: all six Tits equations involve no Grassmann coordinate p_{16}, and e_1, e_6 span a totally singular line. Fix $c \in \mathbb{F}$ with $-c$ a non-square. For $z = (\alpha, \beta) \in \mathbb{F}^2$ let

$$\Pi_z\colon\ X_0 + cX_4 + \alpha X_2 + \beta X_5 = 0, \qquad \Pi = \Pi_{(0,0)},$$

and let S_z be the set of lines of $\mathrm{H}(q)$ contained in Π_z, $S = S_{(0,0)}$. All Π_z contain v.

Lemma 6.1. *For every $z \in \mathbb{F}^2$:*

(a) $\Pi_z \cap \mathrm{Q}(6,q)$ *is an elliptic quadric* $\mathrm{Q}^-(5,q)$.
(b) *Every point of* $\mathrm{H}(q)$ *in* Π_z *lies on exactly one line of* S_z. *So* $|S_z| = q^3 + 1$, *the lines of* S_z *are pairwise opposite, and* $v \in S_z$. (S_z *is a Hermitian spread.*)
(c) *A line of* $\mathrm{H}(q)$ *not in* S_z *meets* Π_z *in exactly one point.*
(d) *The points of* $\mathrm{H}(q)$ *at distance at most* 3 *from* v *are exactly the points of* $\mathrm{Q}(6,q)$ *with* $X_2 = X_5 = 0$. *Among them only the points of* v *lie in* Π_z.
(e) $S_z \cap S_{z'} = \{v\}$ *for* $z \ne z'$.

Proof. (a) On Π_z substitute $X_0 = -cX_4 - \alpha X_2 - \beta X_5$:

$$X_0X_4 + X_1X_5 + X_2X_6 - X_3^2 = (X_1 - \beta X_4)X_5 + X_2(X_6 - \alpha X_4) - (X_3^2 + cX_4^2).$$

This is the sum of two hyperbolic planes and the binary form $X_3^2 + cX_4^2$, which is anisotropic because $-c$ is a non-square. So the form on Π_z is non-degenerate of Witt index 2.

(b) Let $y \in \Pi_z \cap \mathrm{Q}(6,q)$. The lines of $\mathrm{H}(q)$ through y are the lines through y of the totally singular plane π_y. By (a), $\Pi_z \cap \mathrm{Q}(6,q)$ contains no totally singular plane, so $\pi_y \not\subseteq \Pi_z$ and $\pi_y \cap \Pi_z$

is a single line through y; it is the only line of $\mathrm{H}(q)$ on y inside Π_z. As $\mathrm{Q}^-(5,q)$ has $(q+1)(q^3+1)$ points, $|S_z| = q^3+1$. Two lines of S_z are not concurrent by uniqueness. If two lines of S_z were at distance 4, a line N of $\mathrm{H}(q)$ would meet both in distinct points; then $N \subseteq \Pi_z$, so $N \in S_z$ is concurrent with them, a contradiction. So they are opposite.

(c) A line not contained in the hyperplane Π_z meets it in one point.

(d) Let $y = ae_1 + be_6$ be a point of v and let u be a point with yu a line of $\mathrm{H}(q)$. The Tits equations $p_{12} = p_{34}$ and $p_{65} = p_{30}$ and the polarity condition give $au_2 = 0$, $bu_5 = 0$ and $au_5 + bu_2 = 0$, hence $u_2 = u_5 = 0$. So the points at distance at most 3 from v satisfy $X_2 = X_5 = 0$. Conversely, $\mathrm{Q}(6,q) \cap \{X_2 = X_5 = 0\}$ is $X_0X_4 = X_3^2$, a cone with vertex v over a conic; it has $(q+1)+(q+1)q^2$ points, which is the number of points at distance at most 3 from v ($q+1$ on v, and q further points on each of the $q(q+1)$ other lines meeting v). If such a point also lies in Π_z, then $X_0 = -cX_4$ and $-cX_4^2 = X_3^2$, so $X_0 = X_3 = X_4 = 0$ and the point is on v.

(e) Let $m \in S_z \cap S_{z'}$, $m \neq v$. On m, $(\alpha - \alpha')X_2 + (\beta - \beta')X_5 = 0$, so the linear map $m \to \mathbb{F}^2$, $X \mapsto (X_2, X_5)$, has a non-trivial kernel, and m has a point with $X_2 = X_5 = 0$. By (d) that point is at distance at most 3 from v. But m and v are opposite by (b), so every point of m is at distance 5 from v. □

6.2. **Delorme's doubling and whiskers.** Let $\mathcal{P}(v)$ be the set of points of v, $\mathcal{L}_v$ the set of lines $\neq v$ meeting v, and $\mathcal{P}_\Pi$ the set of points of $\mathrm{H}(q)$ in Π (the points on lines of S). A point is *free* if it is not in Π and is at distance 5 from v. By Lemma 6.1(d) the points of Π other than those of v are at distance 5 from v, so there are

$$(q+1)(q^4+q^2+1) - (q+1) - q^2(q+1) - q^3(q+1) = q^3(q^2-1)$$

free points.

Definition 6.2 (Whiskers)**.** A *whisker set* is a set W of lines of $\mathrm{H}(q)$, each opposite v and not in S, such that no free point is on two lines of W. It is *exact* if every free point is on exactly one line of W.

A line m opposite v and not in S meets Π in one point, which lies on a unique line of $S \setminus \{v\}$, and its other q points are free. So $|W| \le q^3(q^2-1)/q = q^2(q^2-1)$, with equality if and only if W is exact.

Theorem 6.3. *Let W be a whisker set and $m \ge 2$. Put*

$$\mathcal{A}_2 = \{v\}, \qquad \mathcal{A}_1 = \{v\} \cup \mathcal{P}(v) \cup S, \qquad \mathcal{A}_0 = \mathcal{A}_1 \cup \mathcal{P}_\Pi \cup \mathcal{L}_v \cup W.$$

Then $\mathcal{A}$ satisfies (L0), (L1) *and* (C_6). *Hence $\mathrm{H}(q)^{(m)}_{\mathcal{A}}$ has maximum degree $q+m$, diameter* 6, *and*

$$2(q+1)(q^4+q^2+1) + (m-1)\big(q^4+2q^3+q^2+2q+2+|W|+m(q^3+q+2)+m^2\big)$$

elements.

For $W = \emptyset$ this is Delorme's construction [7, 8]; at $m = 2$ it gives $2(q+1)(q^4+q^2+1)+q^4+4q^3+q^2+4q+10$ elements, the previous table values 8992, 142464, 374452, 842048 at $\Delta = 7, 11, 13, 15$, and at $q = 13$, $m = 3$ the value 884062 at $\Delta = 16$.

Proof. (L1), $i = 1$: the elements incident with v are its points, which are in $\mathcal{A}_1$; a point of v is incident with exactly one element of $\mathcal{A}_2$, a line of $S \setminus \{v\}$ with none. (L1), $i = 0$: the elements incident with $\mathcal{A}_1$ are the points of v, the lines of $\mathcal{L}_v$, and the points on lines of S, all in $\mathcal{A}_0$. A line of $\mathcal{L}_v$ meets v in one point and contains no other point of $\mathcal{A}_1$. A point of $\mathcal{P}_\Pi \setminus \mathcal{P}(v)$ is on exactly one line of S (Lemma 6.1(b)) and not on v. A line of W contains no point of v, being opposite v. (L0): a point outside $\mathcal{A}_0$ is a point off Π. If it is at distance 3 from v, it is on exactly one line of $\mathcal{L}_v$ (two would close a circuit of length at most 6) and on no line of W (that line would be at distance at most 4 from v). If it is free, it is on no line of $\mathcal{L}_v$ and on at most one line of W. A line outside $\mathcal{A}_0$ is not in Π, so it meets Π in one point, and that point is its only point in $\mathcal{A}_0$. (C_6): a line of $\mathcal{L}_v$

contains no point of Π other than its point on v (Lemma 6.1(d)), and a line of W contains exactly one point of Π, not on v. So the components of the substructure on $\mathcal{A}_0$ are

$$\{v\} \cup \mathcal{P}(v) \cup \mathcal{L}_v \qquad \text{and} \qquad \{n\} \cup \mathcal{P}(n) \cup \{w \in W : w \text{ meets } n\} \quad (n \in S \setminus \{v\}).$$

Both are trees of diameter at most 4: the first is the ball of radius 2 about v, the second consists of n, its points, and whiskers hanging from them. Lemma 3.3 gives (C_6). Now Theorem 3.2 applies. The diameter is exactly 6 by (1), because there are opposite elements of level -1: a free point p (level -1) has q^5 opposite points, and only $(q+1)(q^3+1) < q^5$ points lie in $\mathcal{A}_0$. For the count: $\mathcal{A}_2 = \{v\}$; $\mathcal{A}_1$ consists of q^3+1 lines and $q+1$ points, q^3+q+2 elements; and $\mathcal{A}_0$ consists of the q^3+1 lines of S, the q^2+q lines of $\mathcal{L}_v$, the $|W|$ whiskers and the $(q+1)(q^3+1)$ points of Π, which is $q^4+2q^3+q^2+2q+2+|W|$ elements. □

6.3. **An exact whisker set for every odd q.** For a point X of $\mathrm{H}(q)$ in Π, not on v, put

$$\nu(X) = \frac{X_2X_3 - X_4X_5}{cX_2^2 + X_5^2}.$$

This is well defined: the denominator vanishes only when $X_2 = X_5 = 0$ (as $-c$ is a non-square), and by Lemma 6.1(d) such a point of Π lies on v. Let

$$G(a,b) = (c_0a + c_1b)(a^2 + cb^2) - b^3, \qquad F(z) = \frac{G(\alpha,\beta)}{\alpha^2 + c\beta^2} \quad (z = (\alpha,\beta) \neq 0),$$

where $c_0, c_1 \in \mathbb{F}$ are chosen so that G has no zero on $\mathbb{F}^2 \setminus \{0\}$.

Lemma 6.4. *Such c_0, c_1 exist for every odd q.*

Proof. $G(a,0) = c_0a^3$, so we need $c_0 \neq 0$. For $b \neq 0$ put $\alpha = a/b$: $G(a,b) = 0$ if and only if $c_0\alpha + c_1 = 1/(\alpha^2+c)$ (note $\alpha^2 + c \neq 0$). For fixed α this is a line in the (c_0,c_1)-plane with $q-1$ points having $c_0 \neq 0$. The lines for $\alpha = 0$ and $\alpha = 1$ meet at $c_0 = 1/(1+c) - 1/c = -1/(c(1+c)) \neq 0$ (note $c \neq -1$ as $-c$ is a non-square). So the q lines cover fewer than $q(q-1)$ of the points with $c_0 \neq 0$, and some (c_0,c_1) is good. □

Theorem 6.5. *Let q be odd, $-c$ a non-square, and c_0, c_1 as in Lemma 6.4. Then*

$$W = \big\{\, m \in S_z \setminus \{v\} \;:\; z \in \mathbb{F}^2 \setminus \{0\},\ \nu(m \cap \Pi) = F(z) \,\big\}$$

is an exact whisker set: every free point lies on exactly one line of W, and $|W| = q^2(q^2-1)$.

Here $m \cap \Pi$ is a single point not on v: $m \subseteq \Pi_z$, $m \neq v$, so $m \notin S$ by Lemma 6.1(e), and m is opposite v by Lemma 6.1(b).

The proof rests on a complex structure on each hyperplane Π_z and on an identity that it yields. We use the following notation. A point of Π_z, $z = (\alpha,\beta)$, is written $x = (x_1,\dots,x_6)$, meaning $(X_0,\dots,X_6) = (-cx_4 - \alpha x_2 - \beta x_5,\ x_1,\dots,x_6)$. On Π_z the quadratic form is

$$Q_z(x) = x_1x_5 + x_2x_6 - x_3^2 - cx_4^2 - \alpha x_2x_4 - \beta x_4x_5,$$

with polar form B_z. Put $\mathrm{num}(y) = y_2y_3 - y_4y_5$ and $\mathrm{den}(y) = cy_2^2 + y_5^2$, so that $\nu = \mathrm{num}/\mathrm{den}$ on Π. Define the linear map J_z of Π_z by

$$J_zx = \big(-\beta x_3 + \alpha x_4 - x_6,\ x_5,\ -cx_4 - \beta x_5,\ x_3 + \beta x_2,\ -cx_2,\ cx_1 + \alpha\beta x_2 + \alpha x_3 + c\beta x_4 + \beta^2 x_5\big);$$

for $z = 0$ this is $J_0x = (-x_6, x_5, -cx_4, x_3, -cx_2, cx_1)$.

Lemma 6.6 (Complex structure)**.** *For every $z \in \mathbb{F}^2$:*

(a) *$J_z^2 = -c$, $Q_z(J_zx) = cQ_z(x)$ and $B_z(x, J_zx) = 0$.*
(b) *The six differences $p_{12}-p_{34}$, $p_{54}-p_{32}$, $p_{20}-p_{35}$, $p_{65}-p_{30}$, $p_{01}-p_{36}$, $p_{46}-p_{31}$ of the Grassmann coordinates of $x \wedge J_zx$ are equal to $Q_z(x)$ times 1, 0, 0, $-c$, α, $-\beta$ respectively.*
(c) *For every point x of $\mathrm{Q}(6,q) \cap \Pi_z$, the line $\langle x, J_zx\rangle$ is the line of S_z through x.*

(d) $\mathrm{num}(J_0x) = c\,\mathrm{num}(x)$, $\mathrm{den}(J_0x) = c\,\mathrm{den}(x)$, *and the polar forms of* num *and* den *vanish on* (x, J_0x). *Consequently* ν *is constant on every line of* $S \setminus \{v\}$.

Proof. (a), (b) are polynomial identities of degree 2 in x, checked by direct expansion (we also checked them by computer algebra). For $z = 0$, for instance, $Q_0(J_0x) = cx_2x_6 + cx_1x_5 - c^2x_4^2 - cx_3^2 = cQ_0(x)$, and $p_{12} = x_1x_5 + x_2x_6$, $p_{34} = x_3^2 + cx_4^2$, so $p_{12} - p_{34} = Q_0(x)$.

(c) As $-c$ is a non-square, J_z has no eigenvector, so x and J_zx span a line. If x is on the quadric, then $Q_z(x) = 0$, hence by (a) every point of the line is on the quadric, and by (b) its Grassmann coordinates satisfy Tits' equations. So it is a line of $\mathrm{H}(q)$ in Π_z through x, and by Lemma 6.1(b) it is the line of S_z through x.

(d) The first assertions are direct: $\mathrm{num}(J_0x) = x_5(-cx_4) - x_3(-cx_2) = c\,\mathrm{num}(x)$, $\mathrm{den}(J_0x) = cx_5^2 + c^2x_2^2 = c\,\mathrm{den}(x)$, and the polar forms are $x_2y_3 + x_3y_2 - x_4y_5 - x_5y_4$ and $2cx_2y_2 + 2x_5y_5$, which vanish for $y = J_0x$. By (c) every line $n \in S \setminus \{v\}$ is $\{ax + bJ_0x\}$ for any point x of n, and $\mathrm{num}(ax + bJ_0x) = (a^2 + cb^2)\mathrm{num}(x)$, $\mathrm{den}(ax + bJ_0x) = (a^2 + cb^2)\mathrm{den}(x)$. The factor $a^2 + cb^2$ is non-zero because $-c$ is a non-square. So ν is constant on n. □

In particular the condition defining W depends only on the line of S that m meets: ν is a function on the Hermitian spread $S \setminus \{v\}$.

Let p be a free point. By Lemma 6.1(d), $(p_2, p_5) \neq (0, 0)$, and since $p \notin \Pi$ we may normalize $p_0 + cp_4 = 1$. Then $p \in \Pi_z$ if and only if z lies on the affine line

$$A_p = \{(\alpha, \beta) : \alpha p_2 + \beta p_5 = -1\} \subseteq \mathbb{F}^2 \setminus \{0\},$$

which has q points and direction $(p_5, -p_2)$. For $z \in A_p$ let $m_p(z)$ be the line of S_z through p (Lemma 6.1(b)) and $X(z) = m_p(z) \cap \Pi$.

Lemma 6.7 (Key identity)**.** *For* $z = (\alpha, \beta) \in A_p$,

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + c\beta^2} = \frac{p_2p_3 - p_4p_5 + \beta\, p_2^2}{cp_2^2 + p_5^2}.$$

In particular Φ_p *is the restriction to* A_p *of an affine function, and its increment along* $(p_5, -p_2)$ *is* $-p_2^3/(cp_2^2 + p_5^2)$. *Explicitly, if* $p_5 \neq 0$ *and* $z = (sp_5,\ -1/p_5 - sp_2)$, *then*

$$\Phi_p(z) = -\frac{p_2^3p_5\, s + p_2^2 - p_2p_3p_5 + p_4p_5^2}{p_5(cp_2^2 + p_5^2)};$$

if $p_5 = 0$ *and* $z = (-1/p_2,\ s)$, *then* $\Phi_p(z) = (p_2s + p_3)/(cp_2)$.

Proof. Put $\Theta = cp_2^2 + p_5^2 \neq 0$ and $\gamma = -\alpha p_5 + c\beta p_2$. Write $p = (p_1, \ldots, p_6)$ in the coordinates of Π_z; then $\alpha p_2 + \beta p_5 = -1$. For $y \in \Pi_z$ the quantity $\lambda(y) = y_0 + cy_4$ equals $-\alpha y_2 - \beta y_5$, and Π is the set where it vanishes. We have $\lambda(p) = 1$ and, by the formula for J_z, $\lambda(J_zp) = -\alpha p_5 + c\beta p_2 = \gamma$. By Lemma 6.6(c), $m_p(z) = \langle p, J_zp\rangle$, so $X(z) = \gamma p - J_zp$. Its coordinates are

$$X_2 = \gamma p_2 - p_5, \quad X_3 = \gamma p_3 + cp_4 + \beta p_5, \quad X_4 = \gamma p_4 - p_3 - \beta p_2, \quad X_5 = \gamma p_5 + cp_2.$$

Direct expansion gives

$$\mathrm{den}(X) = (\gamma^2 + c)\,\Theta, \qquad \mathrm{num}(X) = (\gamma^2 + c)(p_2p_3 - p_4p_5) + \beta\big(2\gamma p_2p_5 + cp_2^2 - p_5^2\big).$$

Solving $\alpha p_2 + \beta p_5 = -1$, $-\alpha p_5 + c\beta p_2 = \gamma$ (determinant Θ) gives $\alpha = -(cp_2 + \gamma p_5)/\Theta$, $\beta = (\gamma p_2 - p_5)/\Theta$, and therefore $\alpha^2 + c\beta^2 = (\gamma^2 + c)/\Theta$. Here $\gamma^2 + c \neq 0$ since $-c$ is a non-square (this also shows $\mathrm{den}(X) \neq 0$). Hence

$$\nu(X) = \frac{p_2p_3 - p_4p_5}{\Theta} + \frac{\beta\,(2\gamma p_2p_5 + cp_2^2 - p_5^2)}{(\gamma^2 + c)\Theta}, \qquad \frac{\beta^3}{\alpha^2 + c\beta^2} = \frac{\beta\,(\gamma p_2 - p_5)^2}{(\gamma^2 + c)\Theta},$$

using $\beta^2\Theta^2 = (\gamma p_2 - p_5)^2$. The sum of the two numerators multiplying $\beta/((\gamma^2 + c)\Theta)$ is $2\gamma p_2p_5 + cp_2^2 - p_5^2 + \gamma^2p_2^2 - 2\gamma p_2p_5 + p_5^2 = (\gamma^2 + c)p_2^2$, which gives the stated formula. Along $(p_5, -p_2)$ the

coordinate β changes by $-p_2$. The two explicit formulas follow by inserting $\beta = -1/p_5 - sp_2$ and $\beta = s$. □

Remark 6.8. The identities in Lemmas 6.6 and 6.7 are polynomial identities with integer coefficients, so they hold in every odd characteristic. We also verified them symbolically, and, for $q = 3, 5, 7$, the construction of W itself on all free points of $\mathrm{H}(q)$.

Proof of Theorem 6.5. Let p be a free point, normalized as above. The lines of $\bigcup_{z\neq 0} S_z \setminus \{v\}$ through p are exactly the lines $m_p(z)$, $z \in A_p$, and these q lines are distinct: a common line would lie in $S_z \cap S_{z'} = \{v\}$. Since $F(z) = c_0\alpha + c_1\beta - \beta^3/(\alpha^2 + c\beta^2)$, we have $m_p(z) \in W$ if and only if

$$\Phi_p(z) - (c_0\alpha + c_1\beta) = 0.$$

The left side is affine on A_p with increment

$$-\frac{p_2^3}{cp_2^2 + p_5^2} - (c_0p_5 - c_1p_2) = -\frac{G(p_5, -p_2)}{cp_2^2 + p_5^2} \neq 0$$

along $(p_5, -p_2)$, so it vanishes at exactly one of the q points of A_p. Hence p is on exactly one line of W. The lines of W are opposite v and not in S, so W is an exact whisker set. □

Corollary 6.9. *For every odd prime power q and every $m \geq 2$,*

$$n_b(q + m, 6) \geq 2(q + 1)(q^4 + q^2 + 1) + (m - 1)\big(2q^4 + 2q^3 + 2q + 2 + m(q^3 + q + 2) + m^2\big).$$

In particular $n_b(q + 2, 6) \geq 2(q + 1)(q^4 + q^2 + 1) + 2q^4 + 4q^3 + 4q + 10$ and $n_b(q + 3, 6) \geq 2(q + 1)(q^4 + q^2 + 1) + 4q^4 + 10q^3 + 10q + 34$.

By Proposition 3.6 an exact whisker set is the most this family can give: with W exact, every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$, and equality $q|\mathcal{A}_0| + 2c = 2N$ holds ($c = q^3 + 1$ components).

6.4. **Regular versions.** The structure of Theorem 6.3 with W exact has equally many points and lines in $\mathcal{A}_0$, but more lines than points in $\mathcal{A}_1$ and $\mathcal{A}_2$; it has $(m-1)m(q^3 - q + m)$ more lines than points. We remove repeated lines and then add flags.

Lemma 6.10 (Repeated lines)**.** *If two lines of an incidence structure have the same set of points, deleting one of them changes no distance between the remaining elements.*

Proof. A shortest path through the deleted line L can pass through its twin L' instead; it does not already contain L', since then it could be shortened. □

Lemma 6.11 (Completion)**.** *Let $\mathcal{S}$ be a finite incidence structure with $\Delta(\mathcal{S}) \leq \Delta$, such that every element has degree at least $\Delta - r$, and $\sum_p(\Delta - \deg p) = \sum_L(\Delta - \deg L) =: T$. If $T > 2r\Delta$, there is a set J of anti-flags such that $(\mathcal{P}, \mathcal{L}, \mathrm{I} \cup J)$ is Δ-regular.*

Proof. Choose a multiset J of point–line pairs in which each element x occurs $\Delta - \deg x$ times (possible as the sums agree). Call an entry *bad* if it is a flag of $\mathcal{S}$ or a repeated pair, and choose J with the fewest bad entries. Suppose (p, L) is bad. In $\mathrm{I} \cup J$, the line L is joined to at most Δ points and each of them occurs in at most r entries of J; likewise for p. So at most $2r\Delta$ entries (p', L') of J have p' joined to L or L' joined to p, and some entry (p', L') has neither. Replacing $(p, L), (p', L')$ by $(p, L'), (p', L)$ removes a bad entry and creates none. So J has no bad entries. □

Theorem 6.12. *For every odd prime power q and every $m \geq 2$ there is a $(q+m)$-regular incidence structure of diameter at most 6 with*

$$2(q + 1)\big(q^4 + q^2 + 1 + (m - 1)(q^3 + m + 1)\big)$$

elements; for $m = 2$ this is $2(q+1)(q^4+q^3+q^2+4)$, and for $m = 3$ it is $2(q+1)(q^4+2q^3+q^2+9)$.

Proof. Start from $\mathrm{H}(q)^{(m)}_{\mathcal{A}}$ with W exact. For $n \in S \setminus \{v\}$ (level 1) and $a \in [m]$, the m versions $n^{a1}, \dots, n^{am}$ all have the point set $\{p^a : p \text{ I } n\}$; for $a, b \in [m]$ the versions $v^{ab1}, \dots, v^{abm}$ all have the point set $\{p^{ab} : p \text{ I } v\}$. So there are $mq^3 + m^2$ classes of m repeated lines. Delete $(m-1)m(q^3-q+m) \le (m-1)(mq^3+m^2)$ lines, at most $m-1$ from each class; by Lemma 6.10 no distance increases, and points and lines are now equal in number. Every point of Π is on exactly one line of S, so each point loses at most $m-1$ lines. In $\mathrm{H}(q)^{(m)}_{\mathcal{A}}$ every version has degree between $q+1$ and $q+m$; after the deletion this still holds, so every deficit is at most $r = m-1$. The deficit sum T is at least that of the $m|W| = mq^2(q^2-1)$ whisker versions, each of degree $q+1$, so $T \ge (m-1)mq^2(q^2-1) > 2(m-1)(q+m)$. Lemma 6.11 completes the structure; adding flags cannot increase distances. The number of points is $(q+1)(q^4+q^2+1)+(m-1)\big((q+1)(q^3+1)+m(q+1)\big)$, which gives the formula. □

At $q = 5, 9, 11, 13$ and $m = 2$, Theorem 6.12 gives 9348, 147500, 386328, 866068 elements, and at $q = 13$, $m = 3$ it gives 927724. For these parameters the number of elements exceeds $M_b(q+m, 5)$, so the diameter is exactly 6.

7. Orders

At $\Delta = q+2$ the gap between the Moore bound and a generalized D-gon of order (q, q) is

$$M_b(q+2, D) - M_b(q+1, D) = 2\sum_{i=1}^{D-1}\big((q+1)^i - q^i\big) = 2(D-1)q^{D-2} + O(q^{D-3});$$

it is $4q+4$ for $D = 3$ and $6q^2+10q+6$ for $D = 4$. Doubling cannot close this gap. By Proposition 3.6, a doubled set satisfies $|\mathcal{A}_0| < 2N/q$, where $N = q^{D-1} + O(q^{D-2})$ is the number of points; the higher levels are smaller still, since every element of $\mathcal{A}_1$ has all its $q+1$ neighbors in $\mathcal{A}_0$. So doubling gains at most $2q^{D-2} + O(q^{D-3})$ elements, roughly a fraction $1/(D-1)$ of the gap.

- $D = 3$: the gain is a constant. A pure doubling gains at most 2 (Proposition 3.5); with patches, three hats gain 8 [2] and four hats gain 10 (Theorem 4.2).
- $D = 4$: the gain is of order q^2. Delorme's doubling gains q^2+q+4. Theorem 5.6 uses a different quadrangle, $\mathrm{AS}(q)$, and has $2q^2(q+2) = 2(q+1)(q^2+1) + 2q^2 - 2q - 2$ elements, which has the order of magnitude of the counting bound $2q^2 + O(q)$; Proposition 5.11 limits the shadow-line variant on $\mathrm{Q}(4, q)$ to at most $q+5$ more.
- $D = 6$: the gain is of order q^4. Delorme's doubling gains $q^4 + O(q^3)$; with an exact whisker set (Theorem 6.5) the gain is $2q^4 + 4q^3 + 4q + 10$, and the single-level part attains the counting bound with equality.

Table 2 lists the resulting values in the range $\Delta \le 16$ of the table of largest known bipartite graphs [5], for the degrees where no generalized polygon of order $(\Delta-1, \Delta-1)$ is known.

8. Open questions

(1) *Diameter* 3. Is there a five-hat structure, or any modification of a projective plane of order q with $\Delta = q+2$, diameter 3 and more than $2(q^2+q+6)$ elements?
(2) *Diameter* 4. For odd $q \ge 9$, are there $q^2 - 2q - 2$ shadow lines on $\mathrm{Q}(4, q)$ with one ovoid of clones (Proposition 5.11)? Do weighted 3-ovoids of $\mathrm{W}(q)$ exist for all odd q? Do the 2-ovoids of Proposition 5.8 coincide with known ones [3, 6]?
(3) *Diameter* 6. Is there a balanced doubling of $\mathrm{H}(q)$, with level-1 centers at points as well as lines, that is regular without deleting repeated lines? The Feit–Higman theorem rules out hexagons of order $(q-1, q+1)$, so there is no direct analogue of $\mathrm{AS}(q)$; are there weighted distance-2 ovoids of $\mathrm{H}(q)$ that help?
(4) *Diameter* 8. The Ree–Tits octagon of order $(2, 4)$ has the shape $(q-1, q+1)$ with $q = 3$; a weighted 2-ovoid in it would give a 5-regular structure of diameter 8 with 5850 elements.

(Δ, D)	q	$M_b(\Delta, D)$	table [5]	new	regular	construction
$(11, 3)$	9	222	190	**192**	192	Thm. 4.2, 4.7
$(13, 3)$	11	314	274	**276**	276	Thm. 4.2, 4.7
$(15, 3)$	13	422	374	**376**	376	Thm. 4.2, 4.7
$(7, 4)$	5	518	346	**359**	$350^{\dagger}$	Thm. 5.4, Q$(4, 5)$, 21 shadow lines*
$(11, 4)$	9	2222	1734	**1782**	1782	Cor. 5.7, AS(9)
$(13, 4)$	11	3770	3064	**3146**	3146	Cor. 5.7, AS(11)
$(15, 4)$	13	5910	4946	**5070**	5070	Cor. 5.7, AS(13)
$(16, 4)$	13	7232	5134	**5440**	5440	Cor. 5.10, Q$(4, 13)$, Prop. 5.8
$(7, 6)$	5	18662	8992	**9592**	9348	Cor. 6.9, Thm. 6.12
$(11, 6)$	9	222222	142464	**148944**	147500	Cor. 6.9, Thm. 6.12
$(13, 6)$	11	542906	374452	**388972**	386328	Cor. 6.9, Thm. 6.12
$(15, 6)$	13	1158390	842048	**870440**	866068	Cor. 6.9, Thm. 6.12
$(16, 6)$	13	1627232	884062	**940846**	927724	Cor. 6.9, Thm. 6.12, $m = 3$

TABLE 2. Values for $\Delta \leq 16$ with $\Delta - 1$ not a prime power. "Table" is the value listed in [5] (consulted October 2026; for $D = 3$ these are the values of [2], for $D = 4$ and $D = 6$ those of Delorme [7, 8]). "New" is the largest structure in this paper; "regular" the largest Δ-regular one. *An explicit example, not a member of a proved infinite family (data: `research/gq4x/graphs/q4_share_q5_n359.json`). †Corollary 5.7 at $q = 5$ (data: `research/gq4x/graphs/as_w2_q5_n350.json`). The cells $(7, 3)$ and $(16, 3)$ are not improved: Theorem 4.2 gives $72 < 80$ at $(7, 3)$. All structures in the table were also built explicitly and their incidence graphs checked by computer.

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